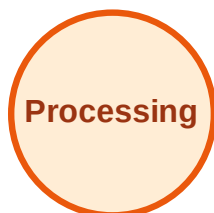


BFS Traversal

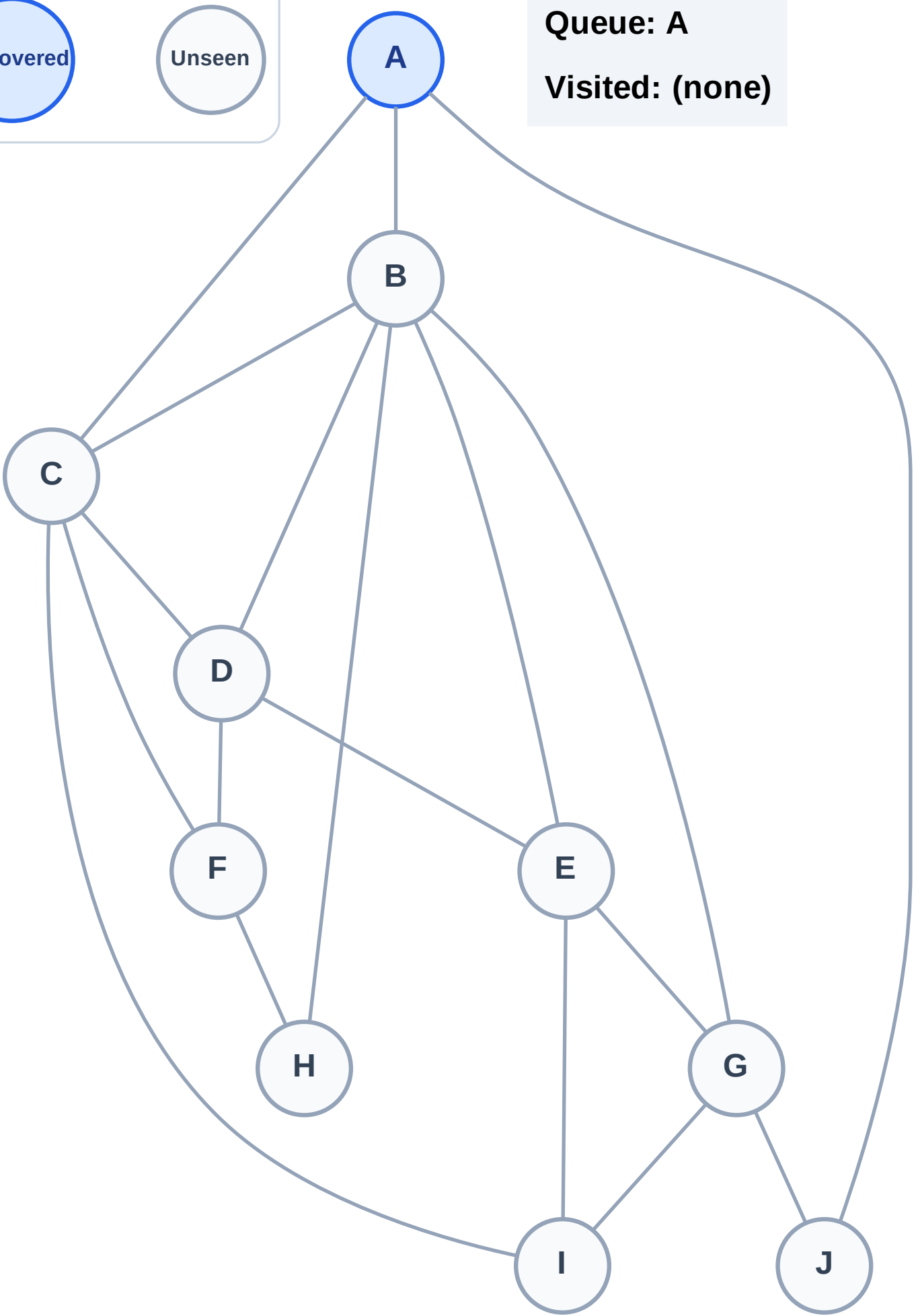
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

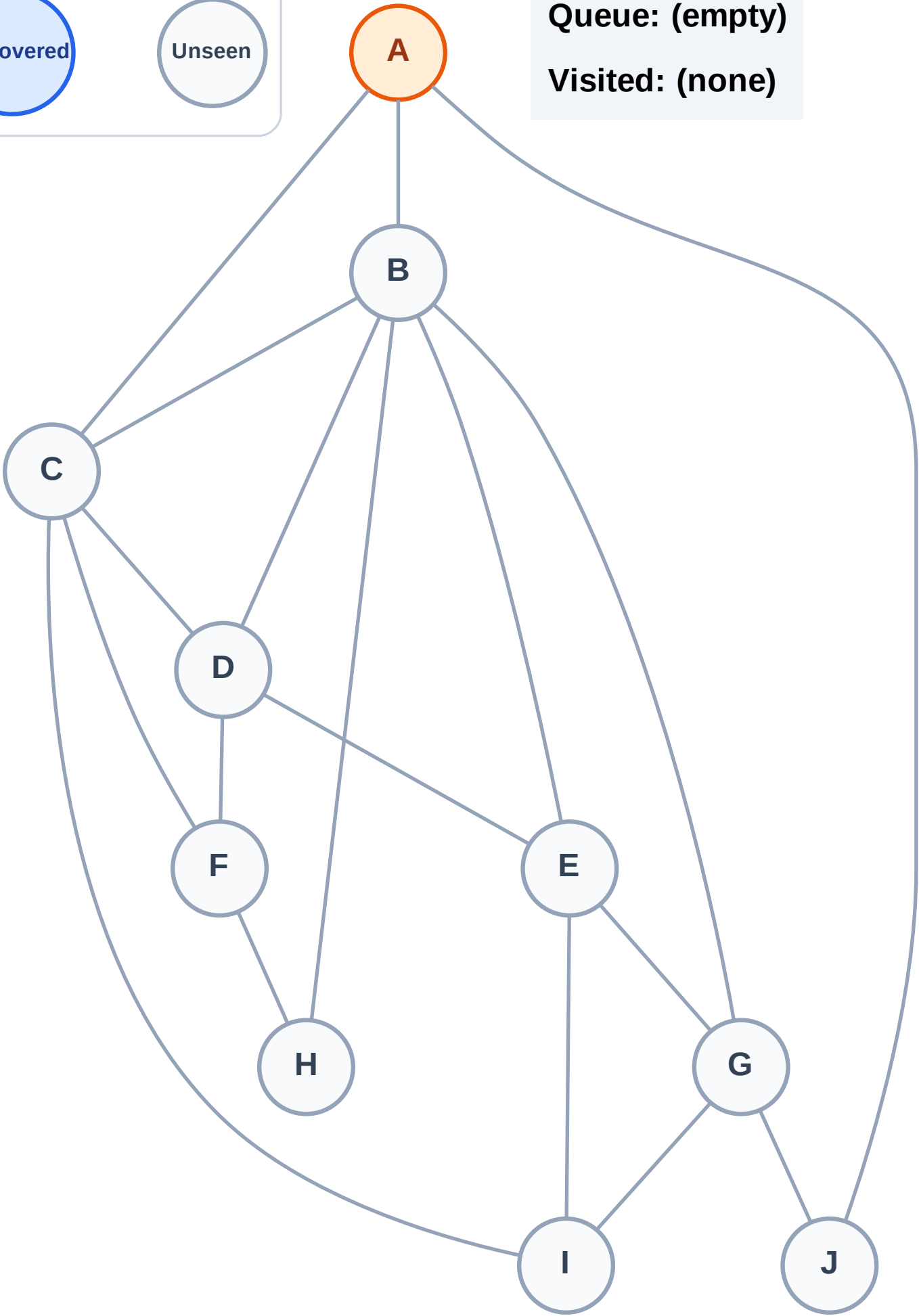
Complete

Processing

Discovered

Unseen

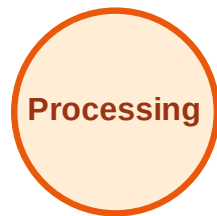
Queue: (empty)
Visited: (none)



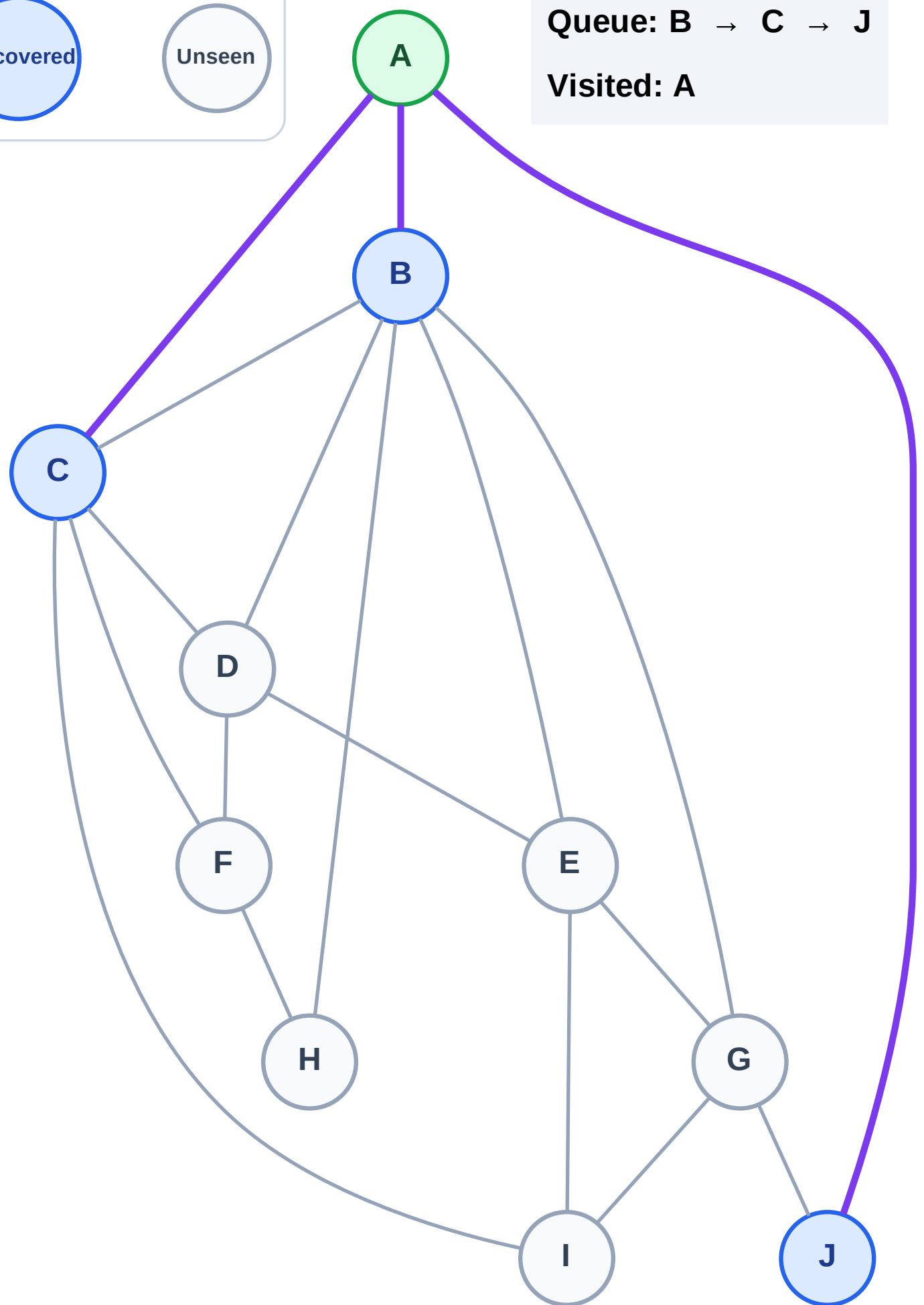
BFS Traversal

Step 3: Discover B, C, J from A

Legend



Queue: B → C → J
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

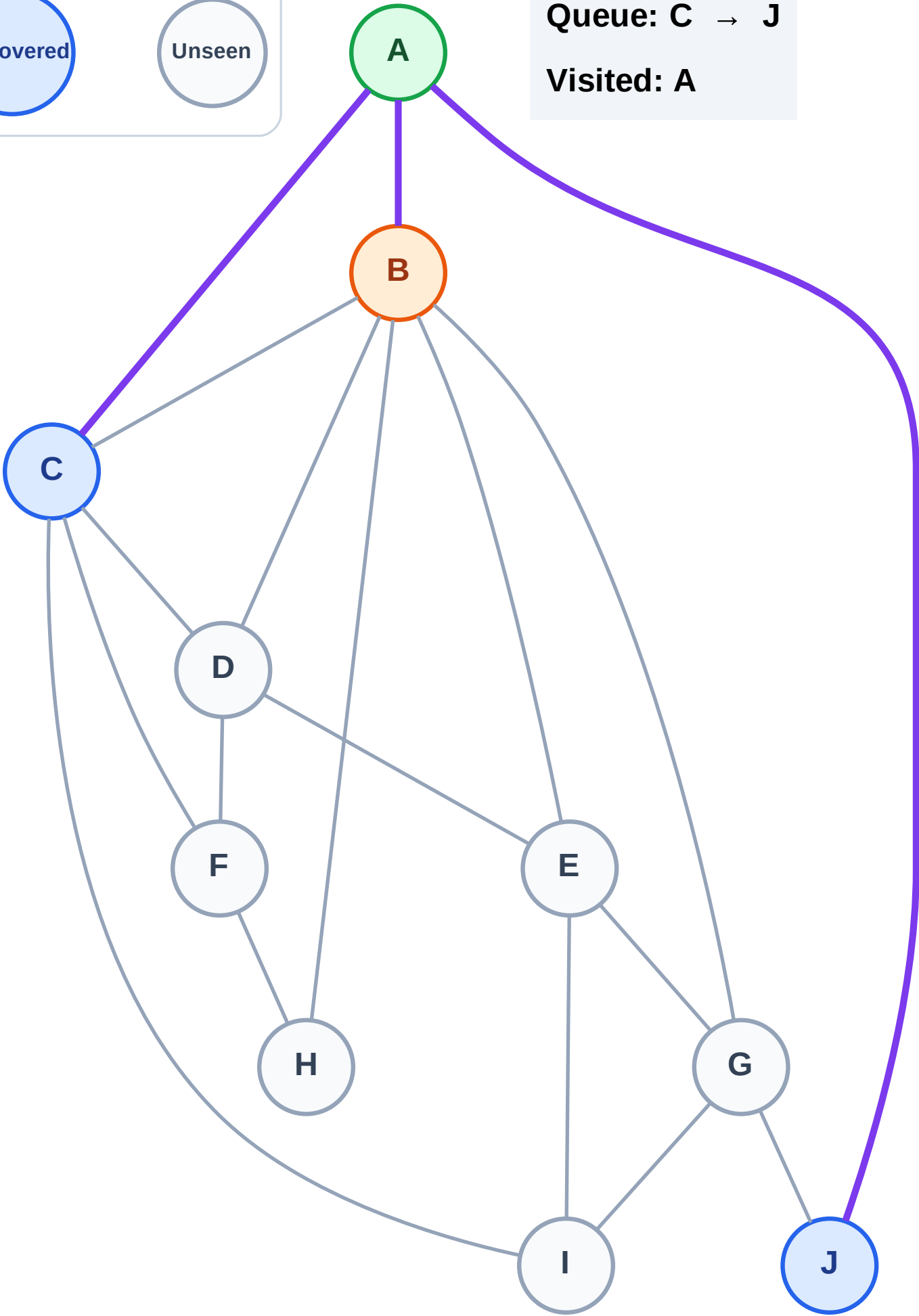
Processing

Discovered

Unseen

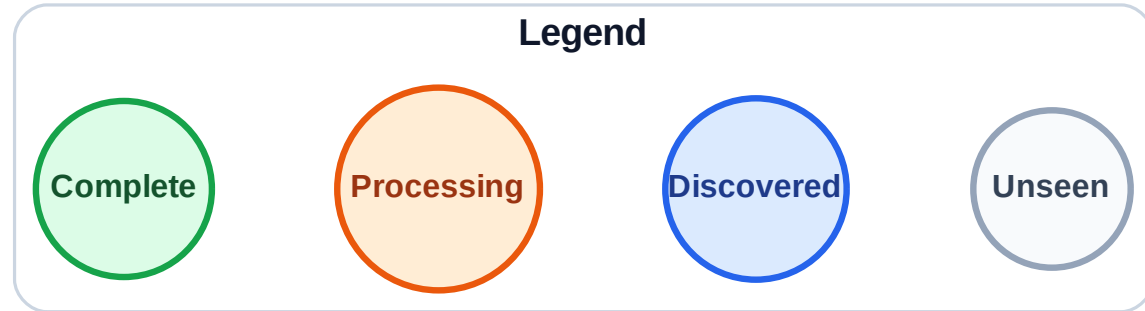
Queue: C → J

Visited: A



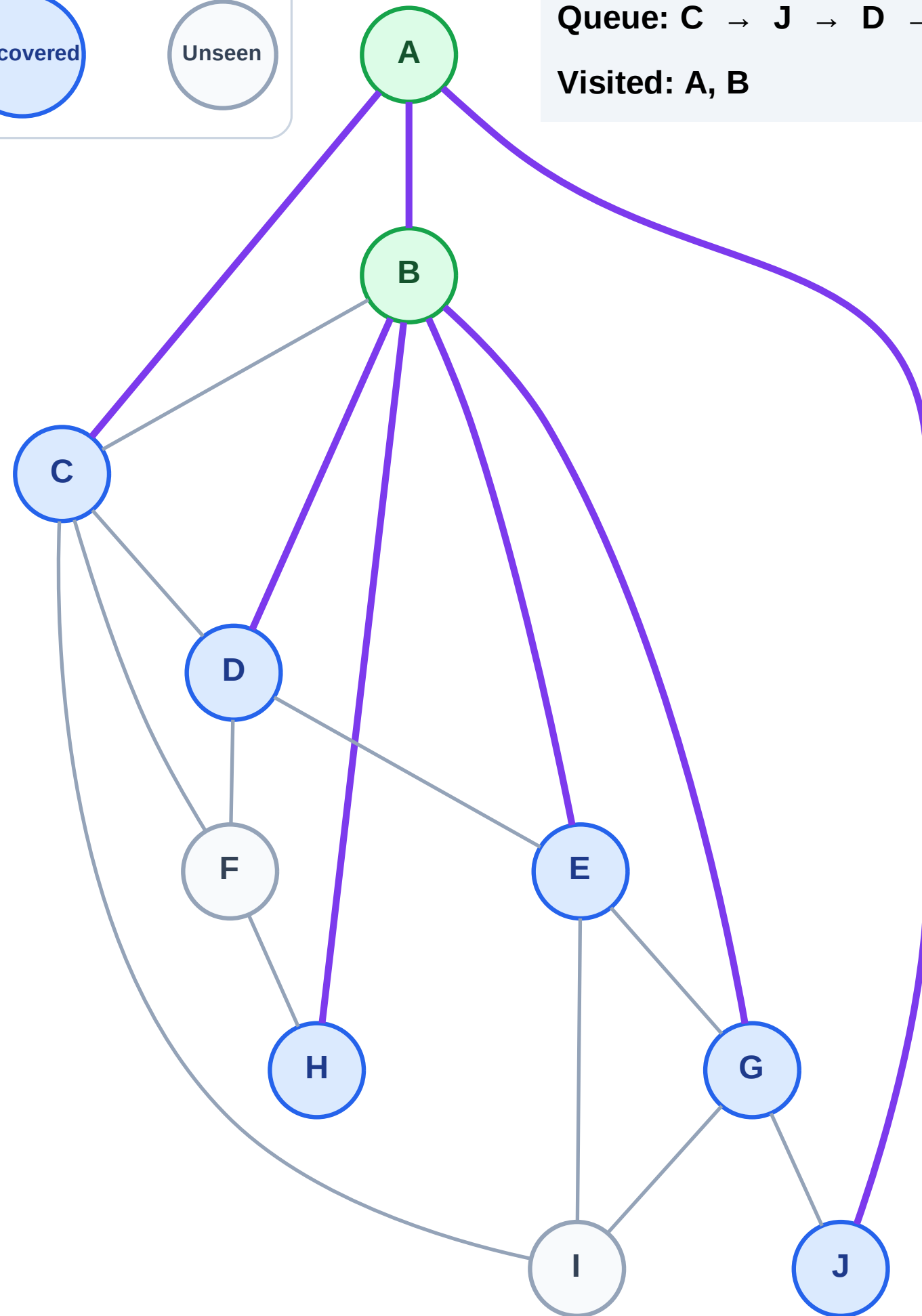
BFS Traversal

Step 5: Discover D, E, G, H from B



Queue: C → J → D → E → G → H

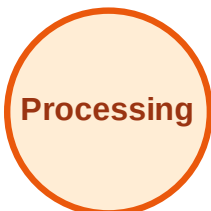
Visited: A, B



BFS Traversal

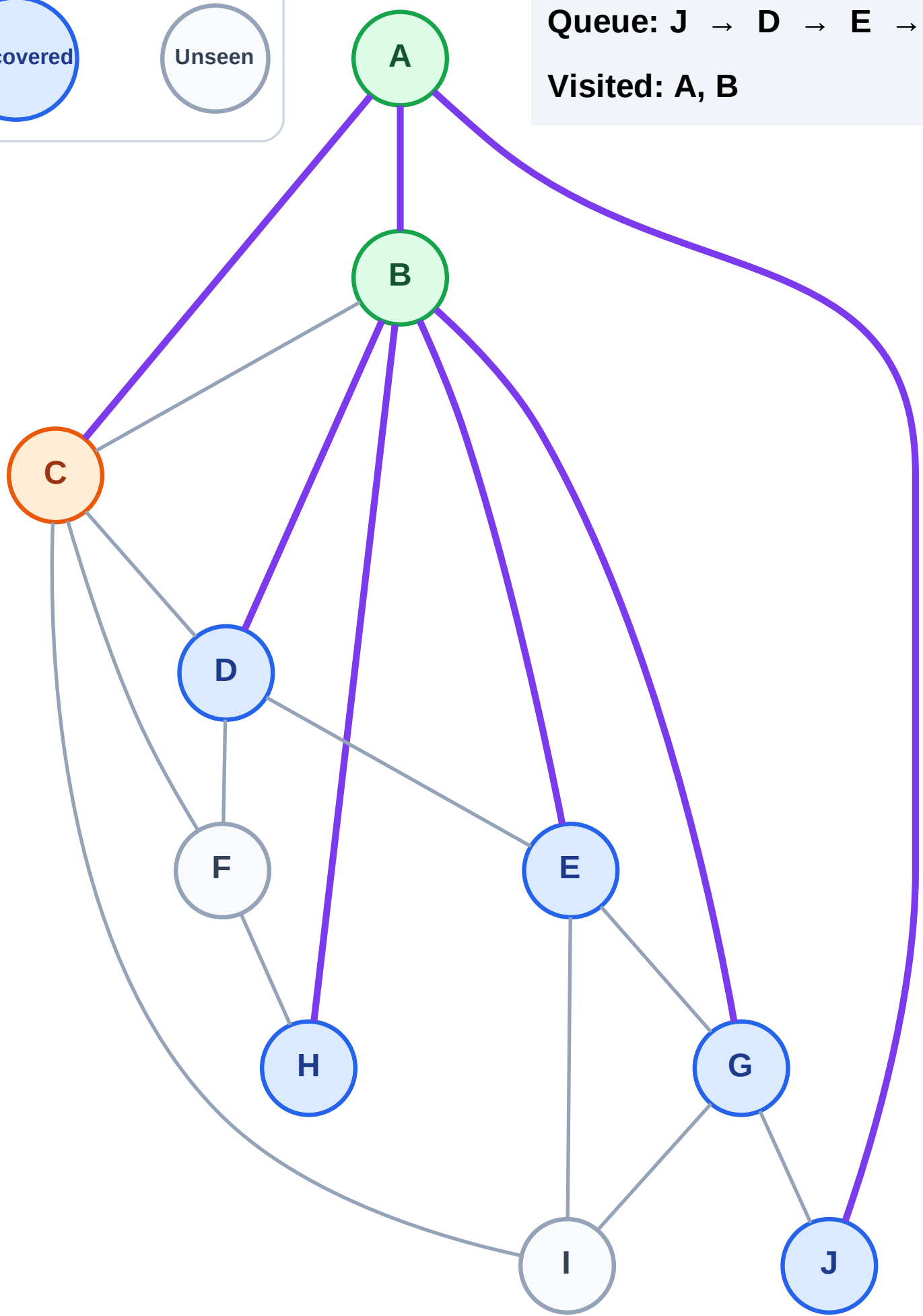
Step 6: Process node C

Legend



Queue: J → D → E → G → H

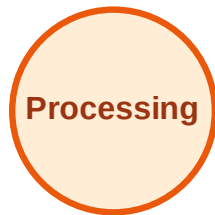
Visited: A, B



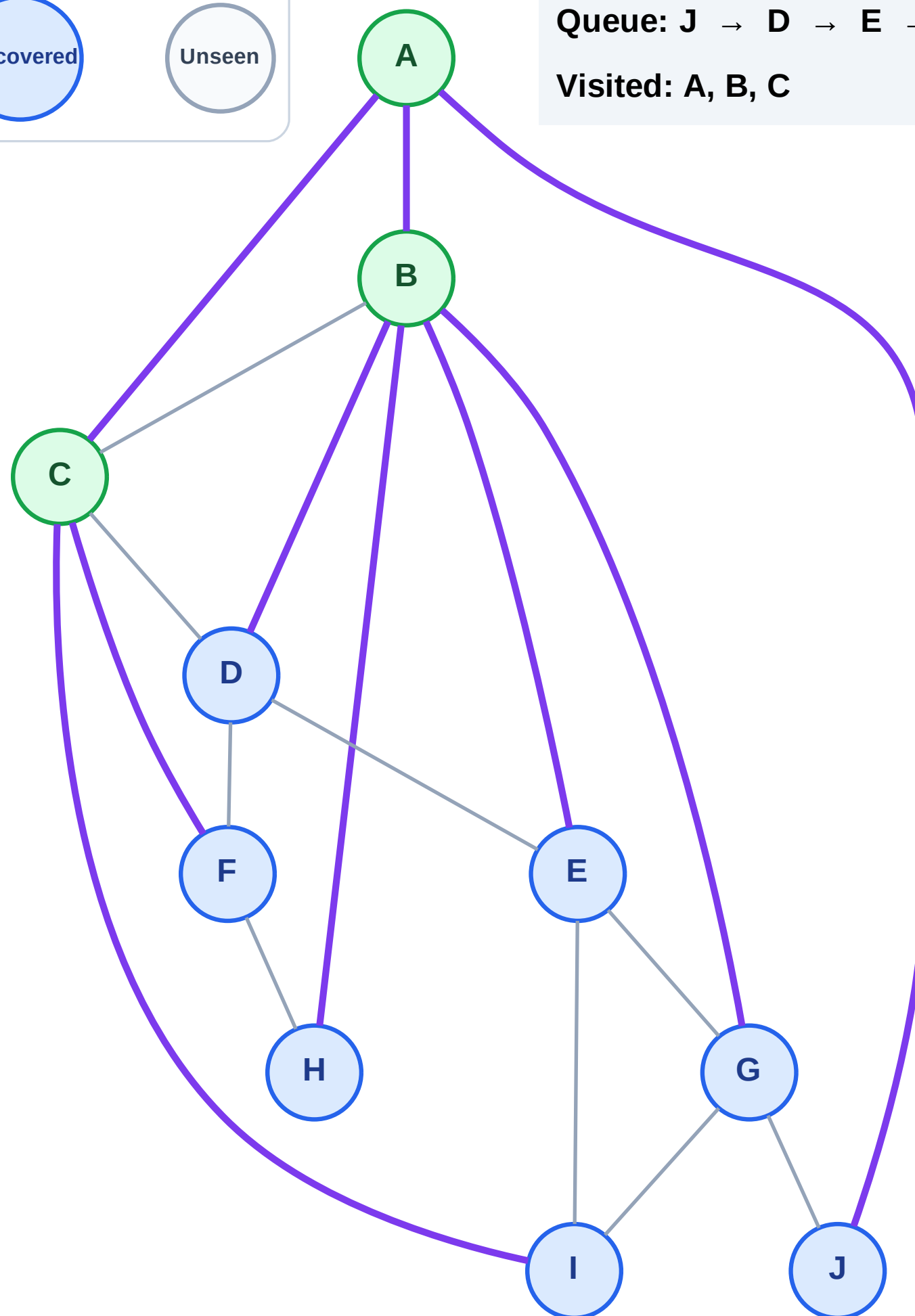
BFS Traversal

Step 7: Discover F, I from C

Legend



Queue: J → D → E → G → H → F → I
Visited: A, B, C



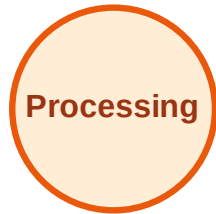
BFS Traversal

Step 8: Process node J

Legend



Complete



Processing



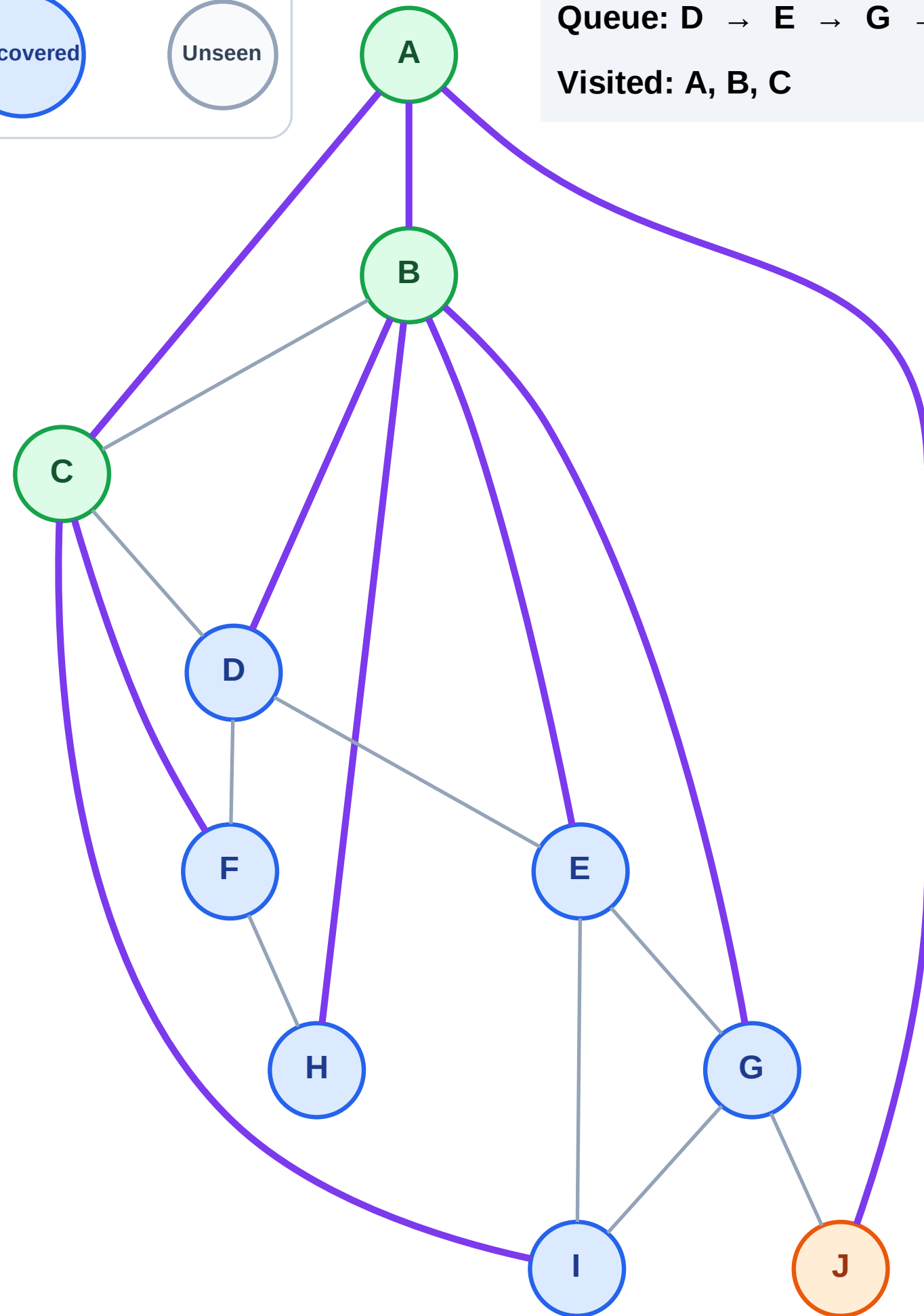
Discovered



Unseen

Queue: D → E → G → H → F → I

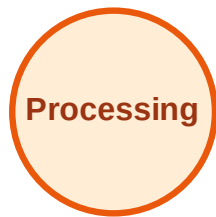
Visited: A, B, C



BFS Traversal

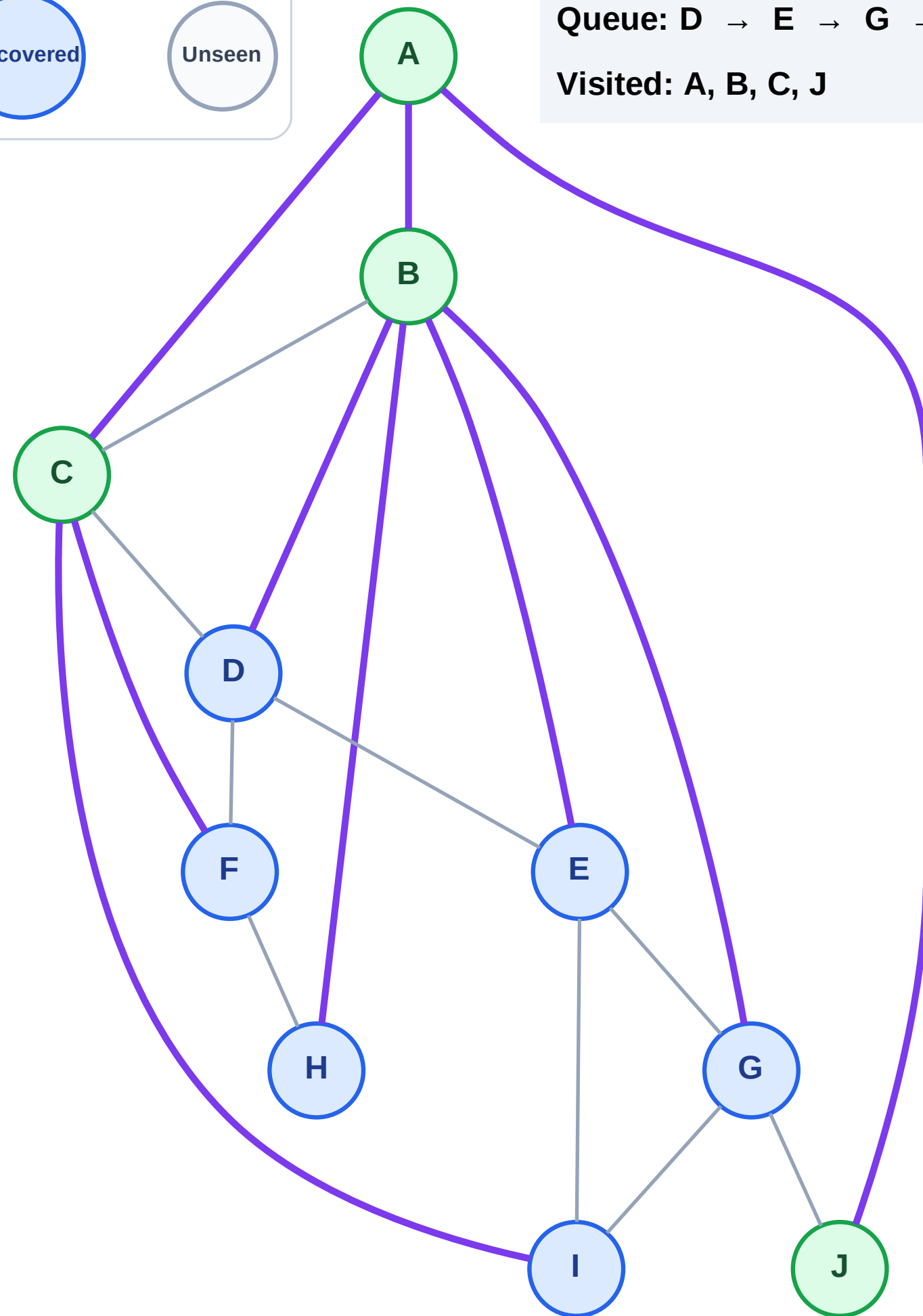
Step 9: No new neighbors from J

Legend

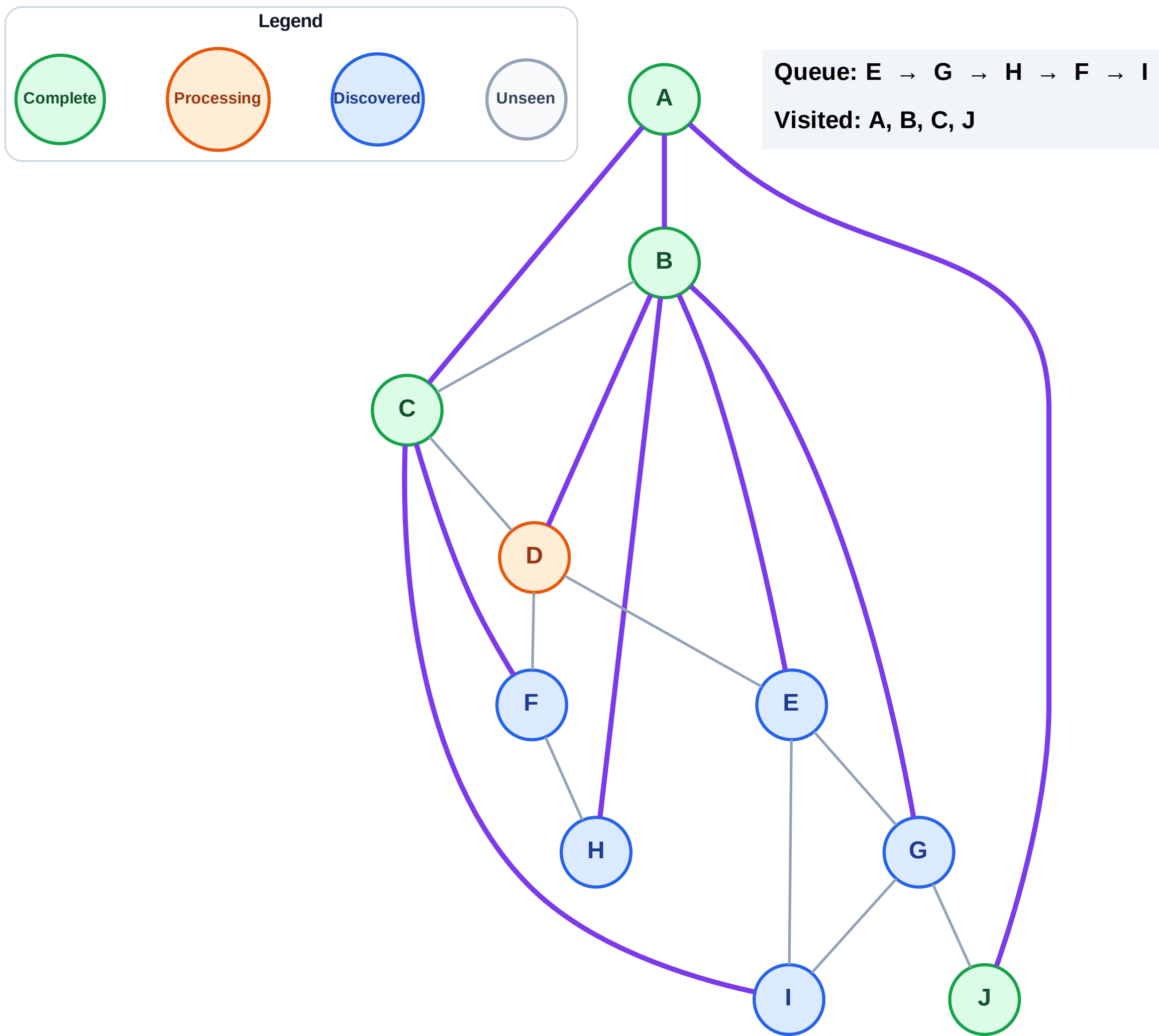


Queue: D → E → G → H → F → I

Visited: A, B, C, J



Step 10: Process node D



BFS Traversal

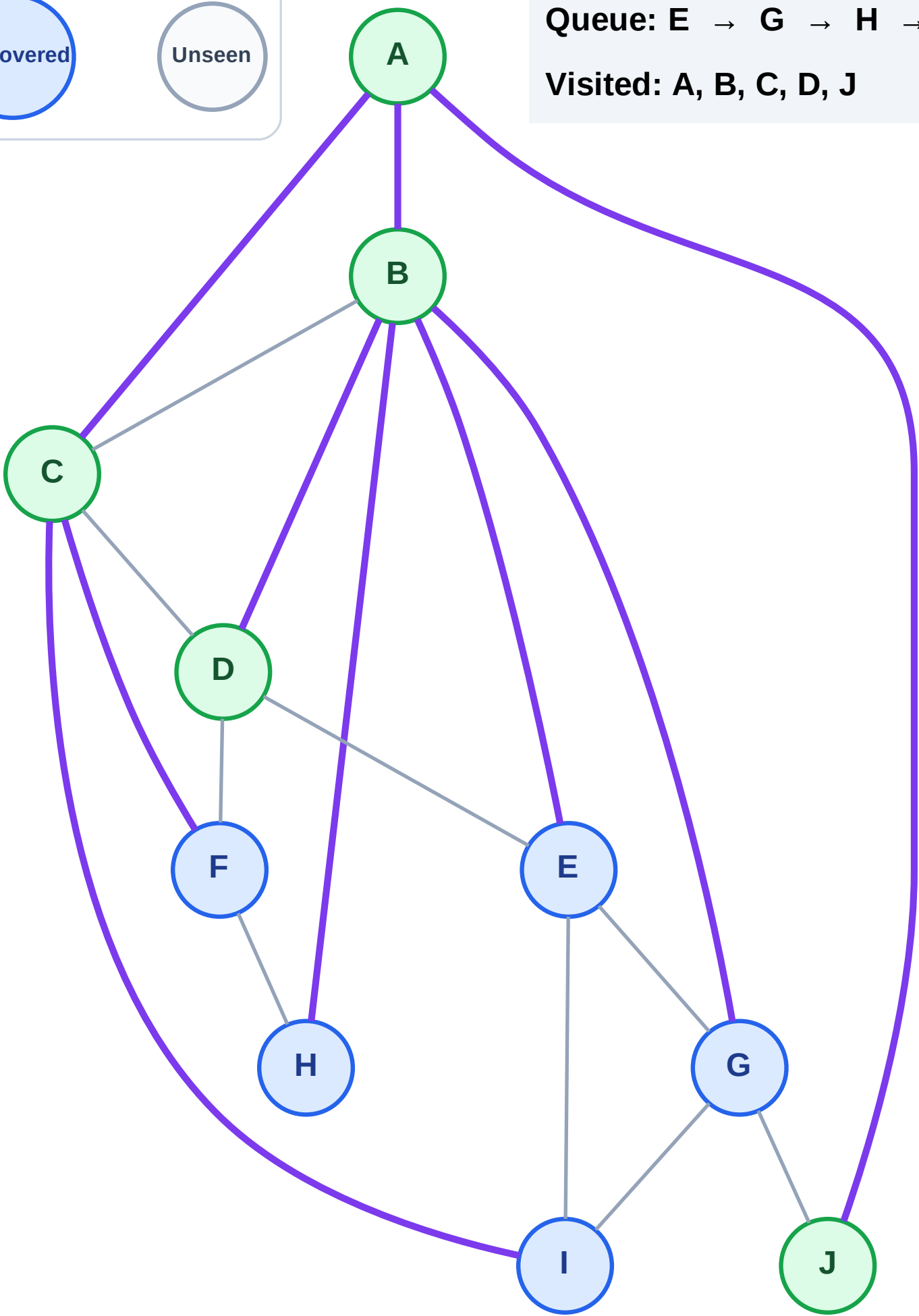
Step 11: No new neighbors from D

Legend



Queue: E → G → H → F → I

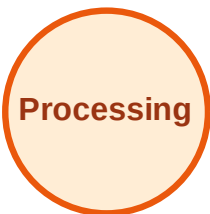
Visited: A, B, C, D, J



BFS Traversal

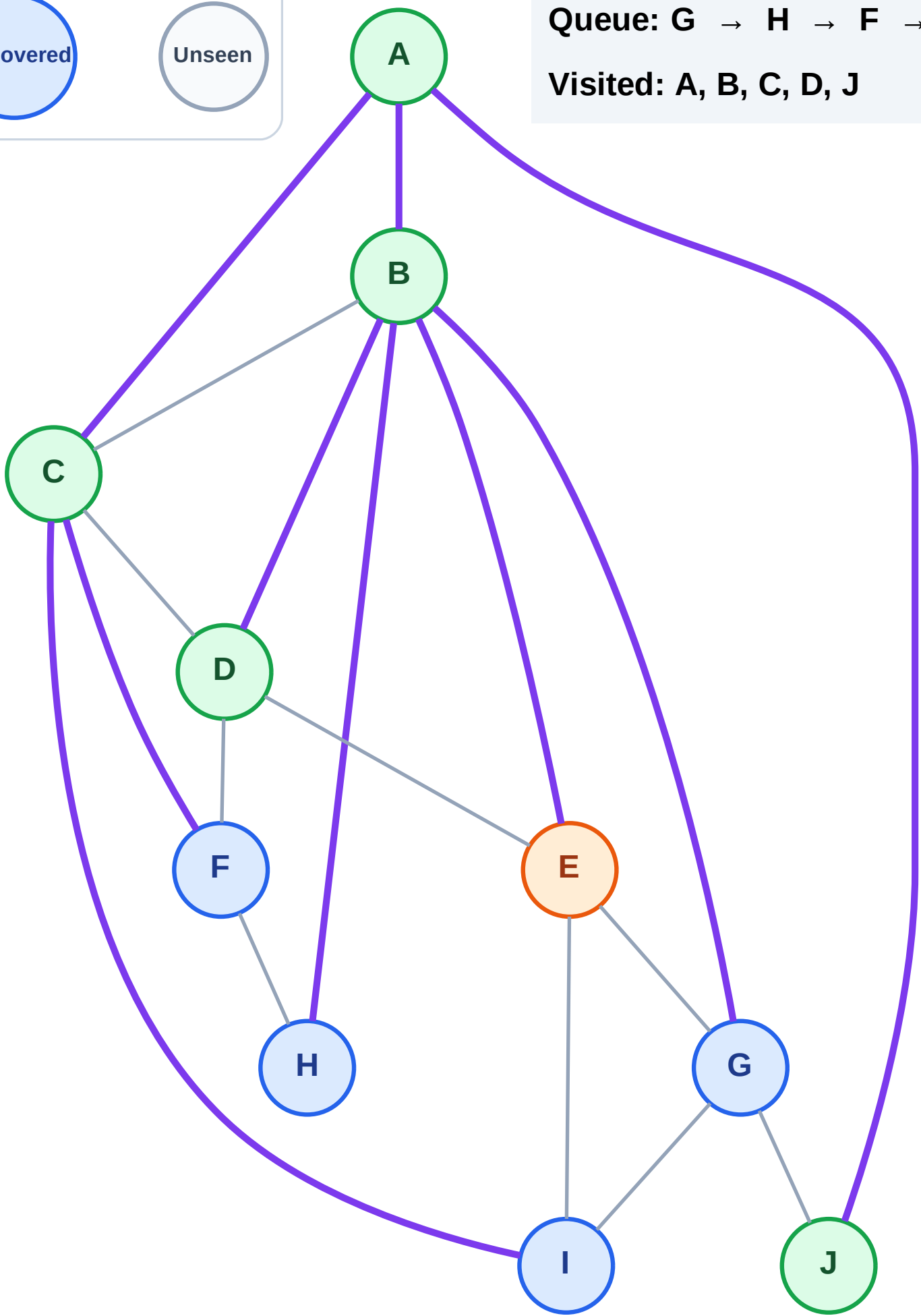
Step 12: Process node E

Legend



Queue: G → H → F → I

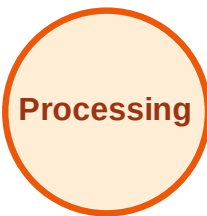
Visited: A, B, C, D, J



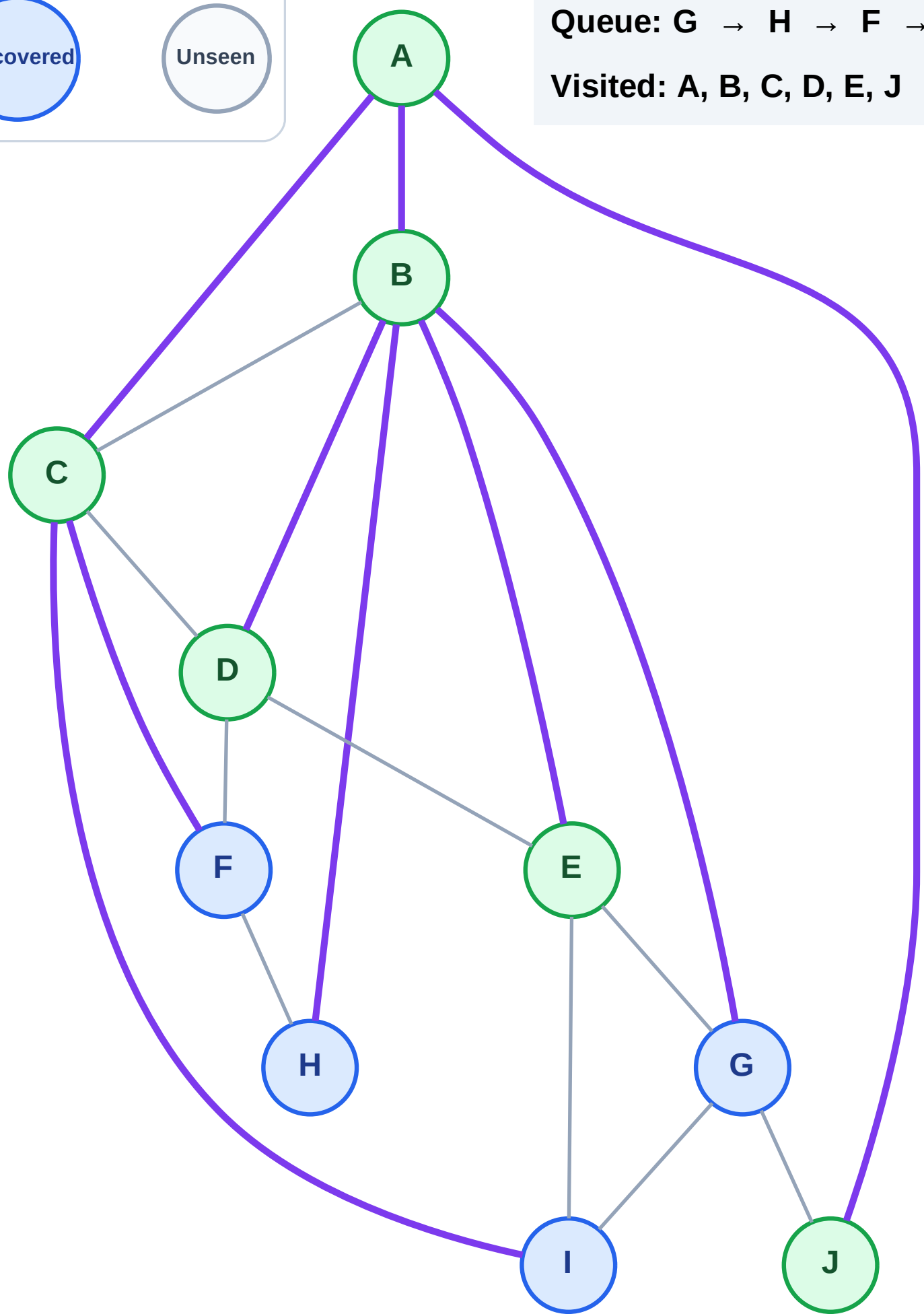
BFS Traversal

Step 13: No new neighbors from E

Legend



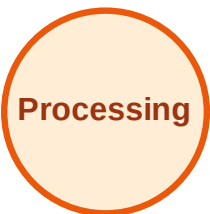
Queue: G → H → F → I
Visited: A, B, C, D, E, J



BFS Traversal

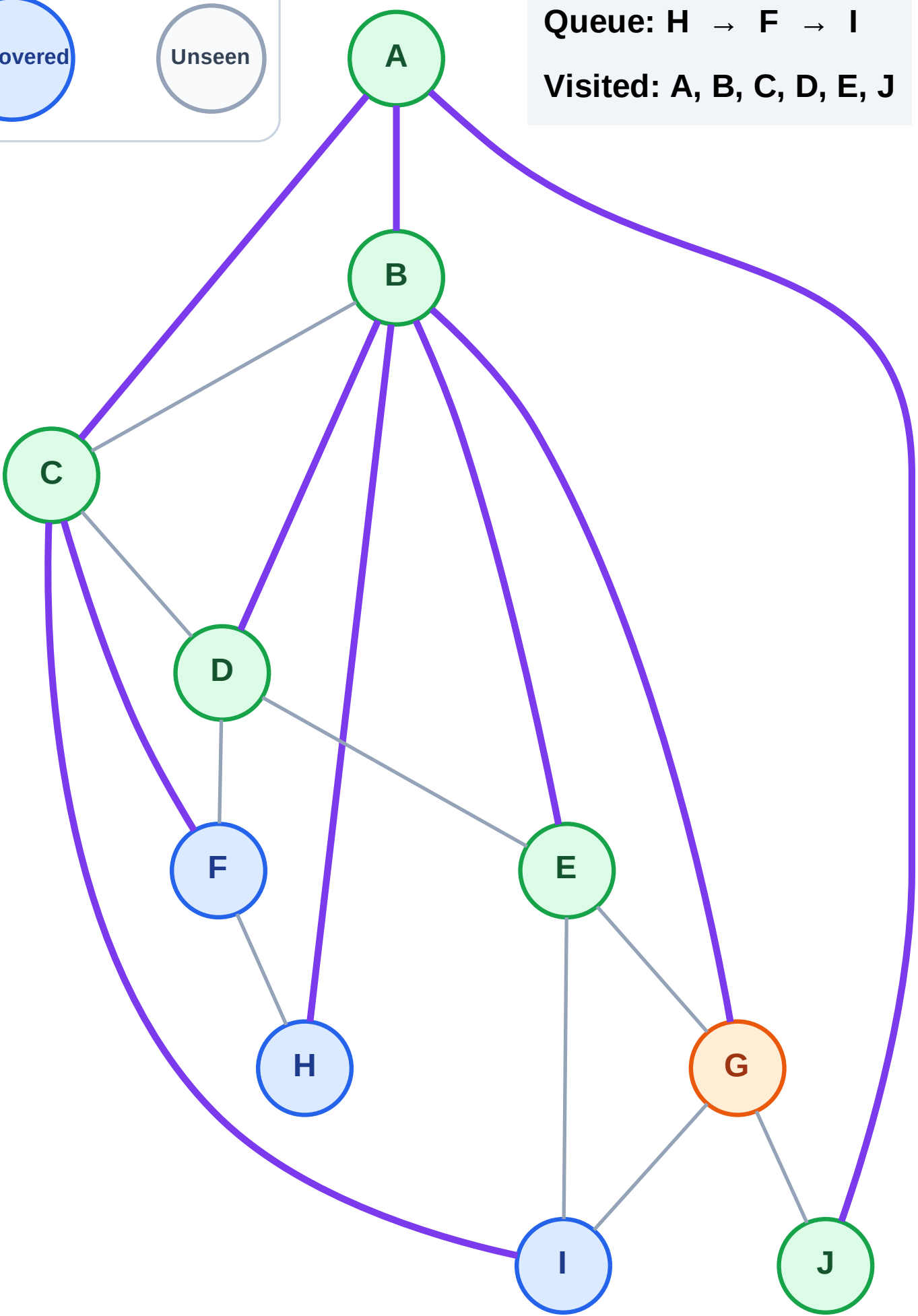
Step 14: Process node G

Legend



Queue: H → F → I

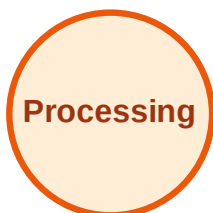
Visited: A, B, C, D, E, J



BFS Traversal

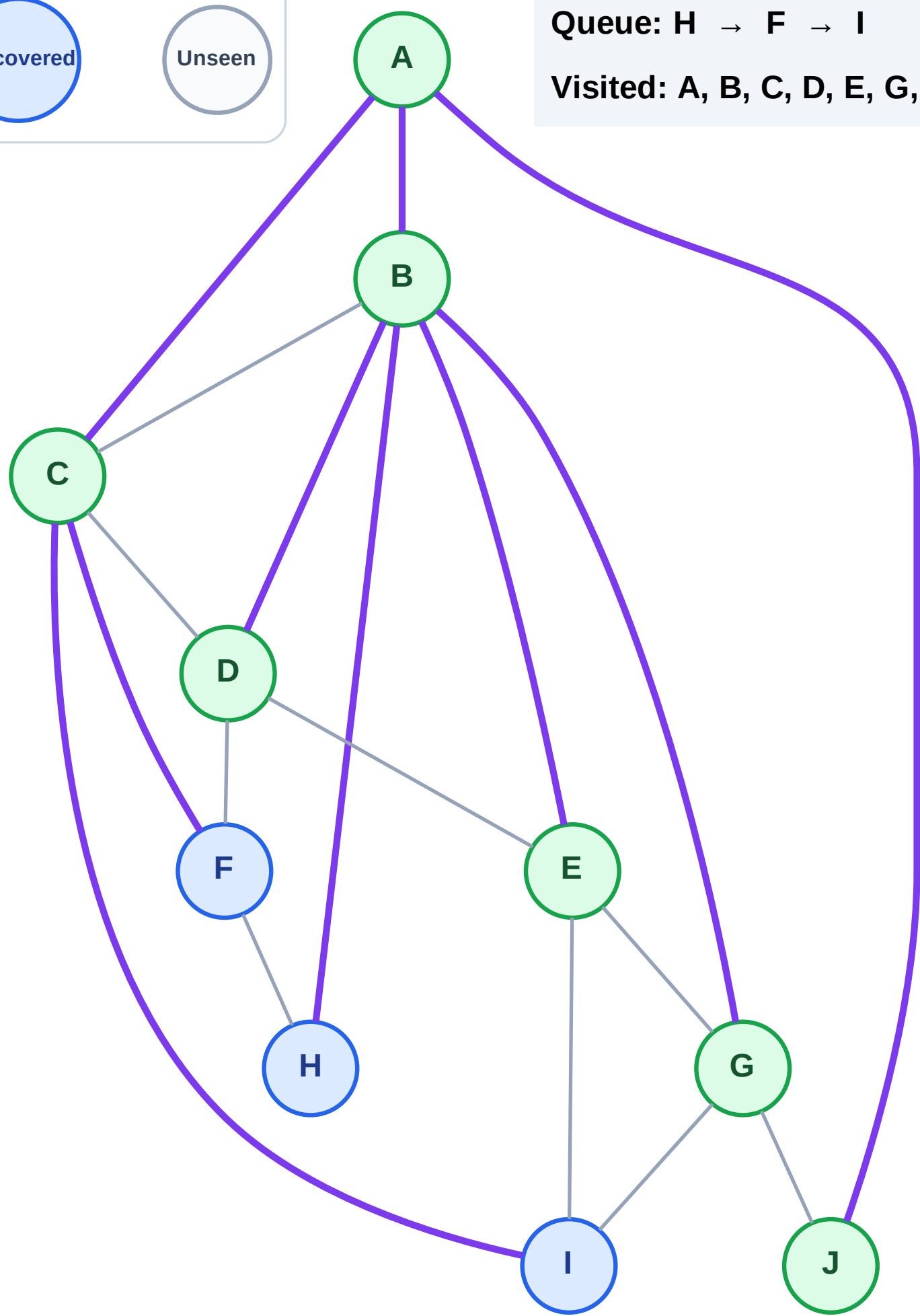
Step 15: No new neighbors from G

Legend



Queue: H → F → I

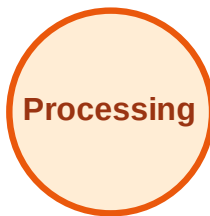
Visited: A, B, C, D, E, G, J



BFS Traversal

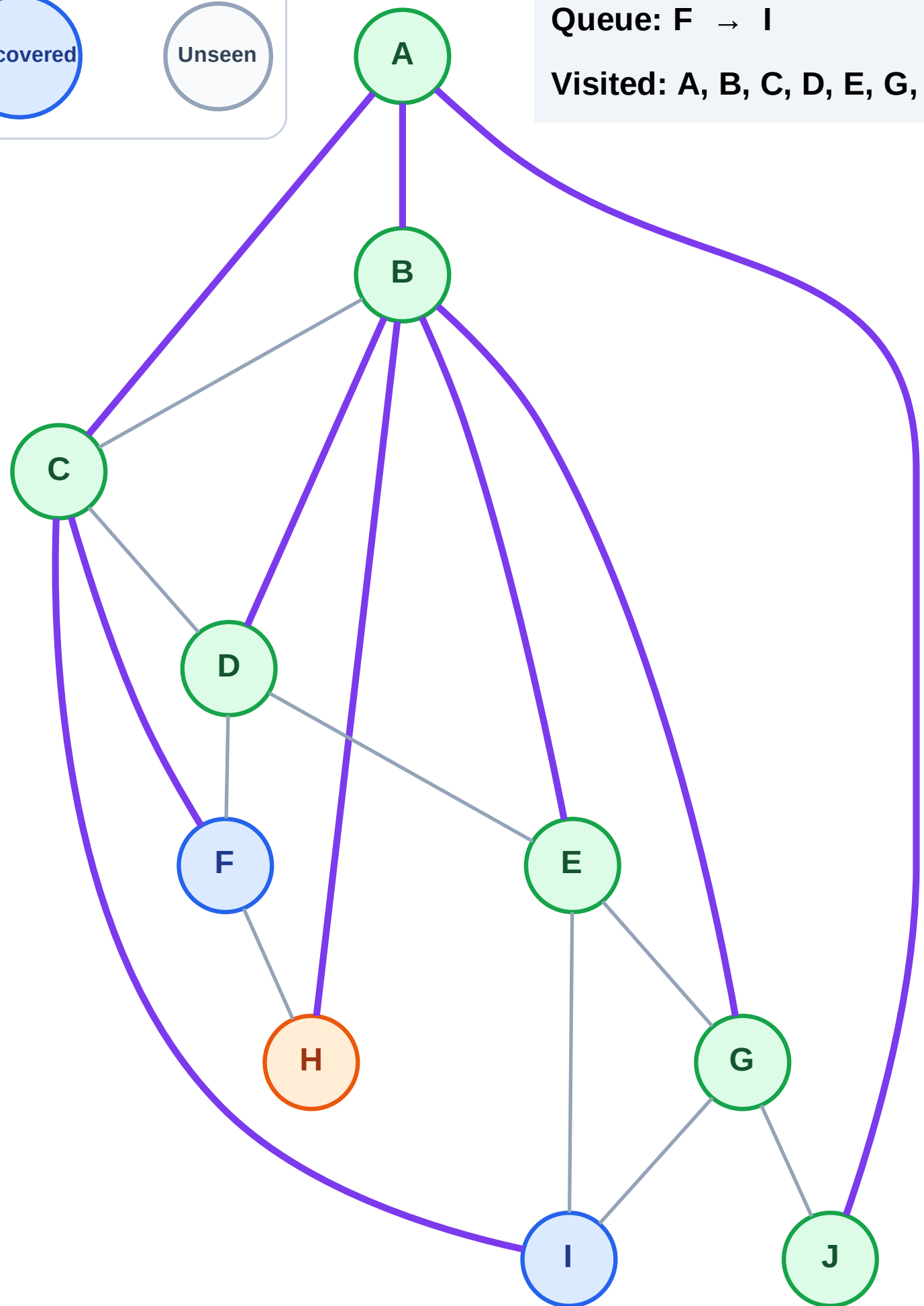
Step 16: Process node H

Legend



Queue: F → I

Visited: A, B, C, D, E, G, J

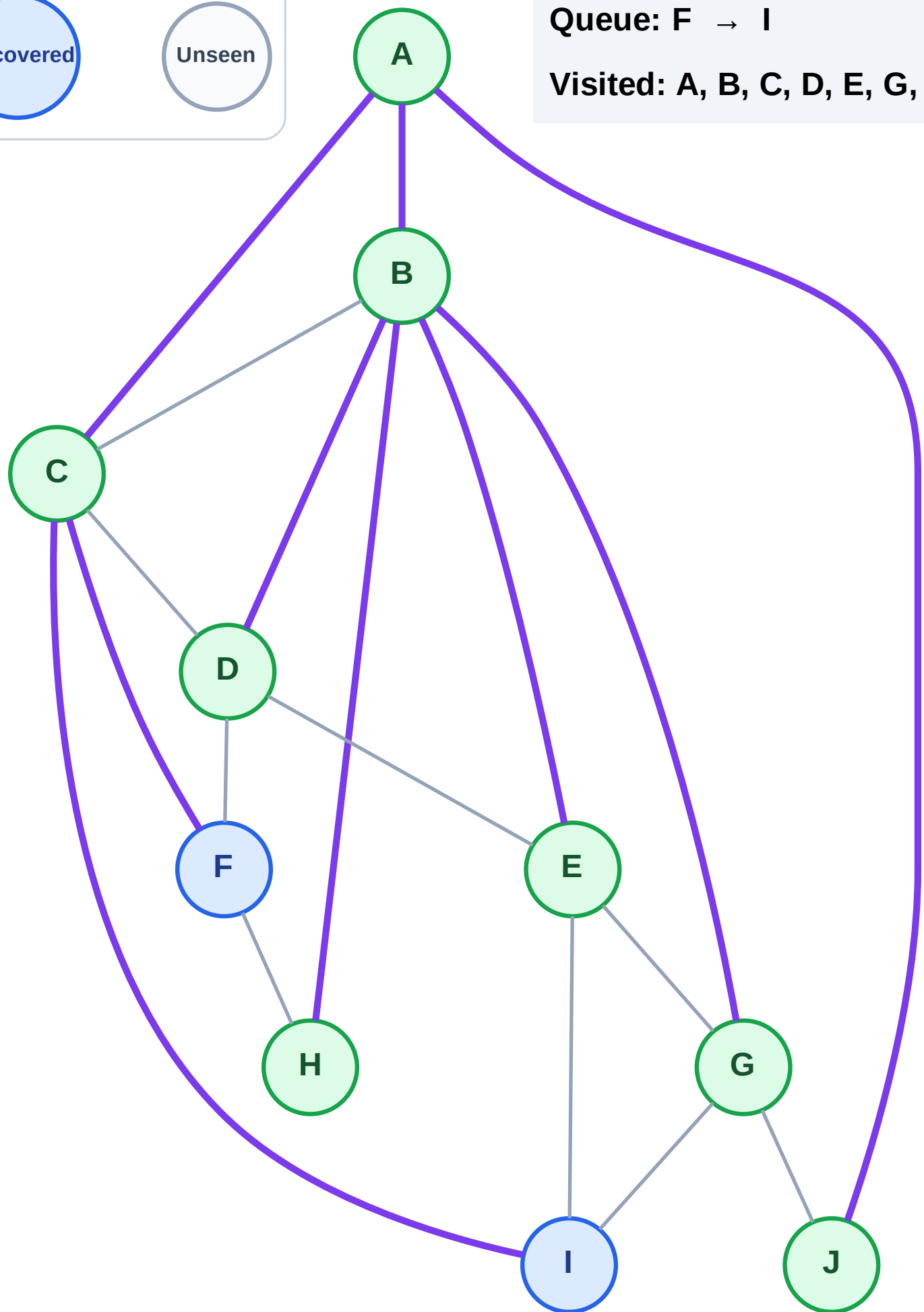


Step 17: No new neighbors from H

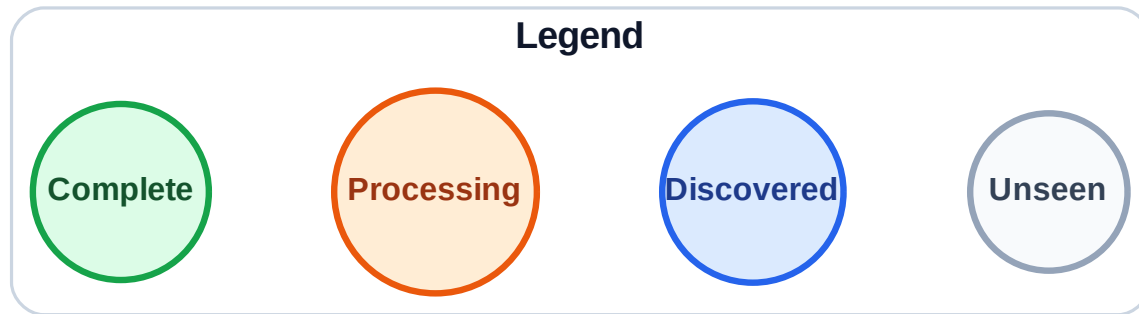
Legend

Unseen

Visited: A, B, C, D, E, G, H, J

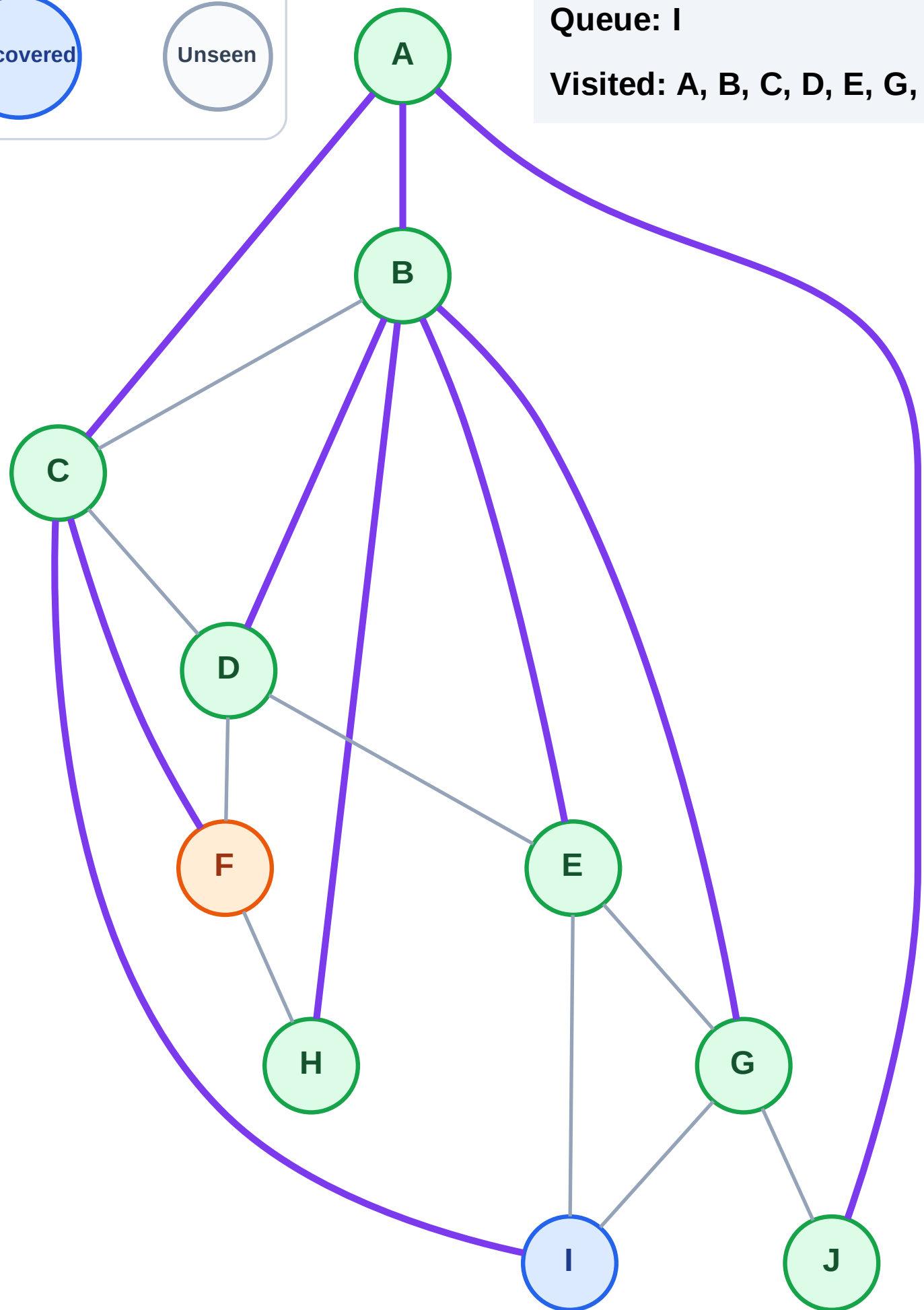


Step 18: Process node F



Queue: I

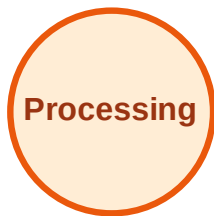
Visited: A, B, C, D, E, G, H, J



BFS Traversal

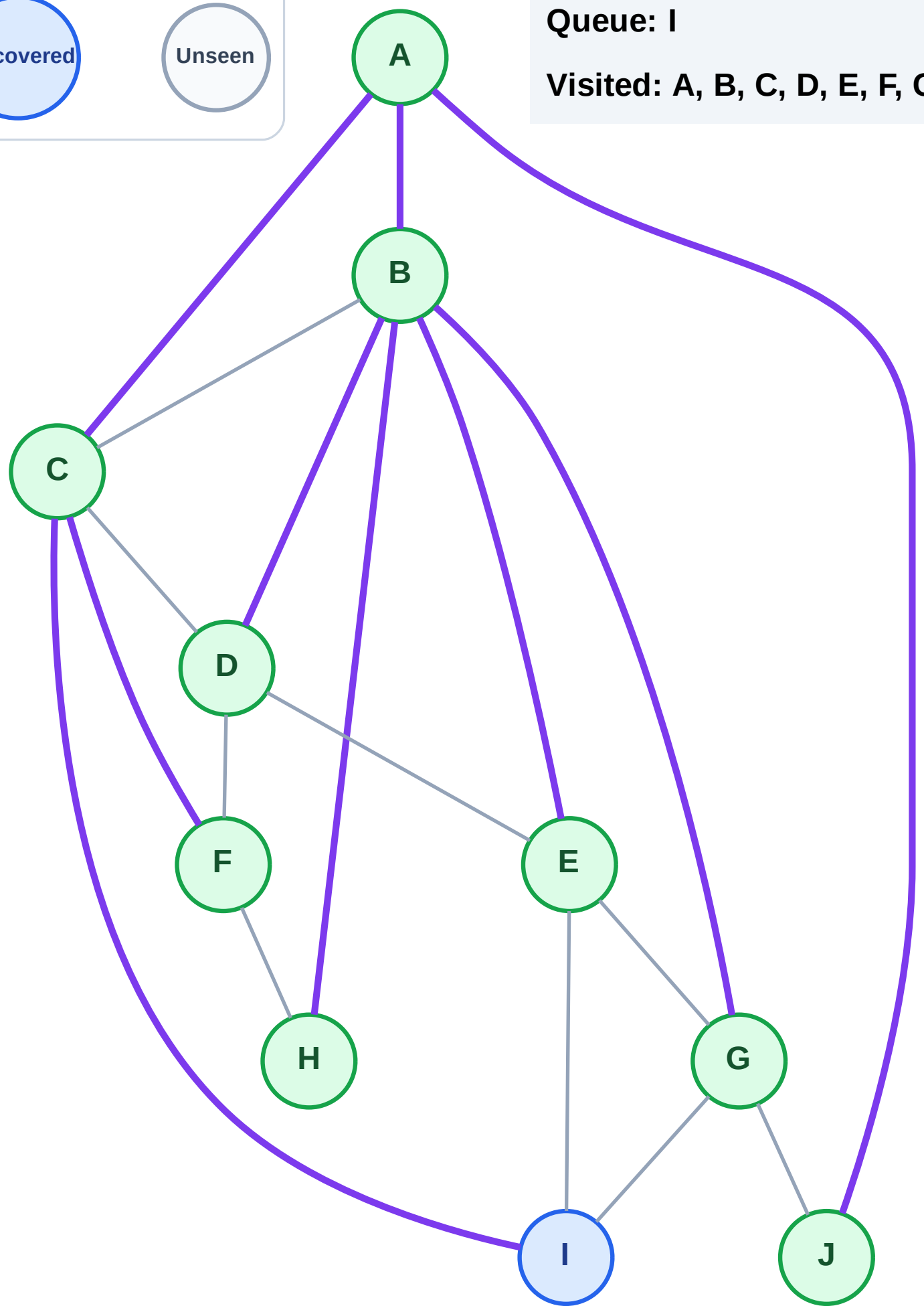
Step 19: No new neighbors from F

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J

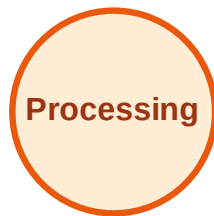


Step 20: Process node I

© 2013 Pearson Education, Inc. or its affiliate(s). All rights reserved.



Complete



Processing



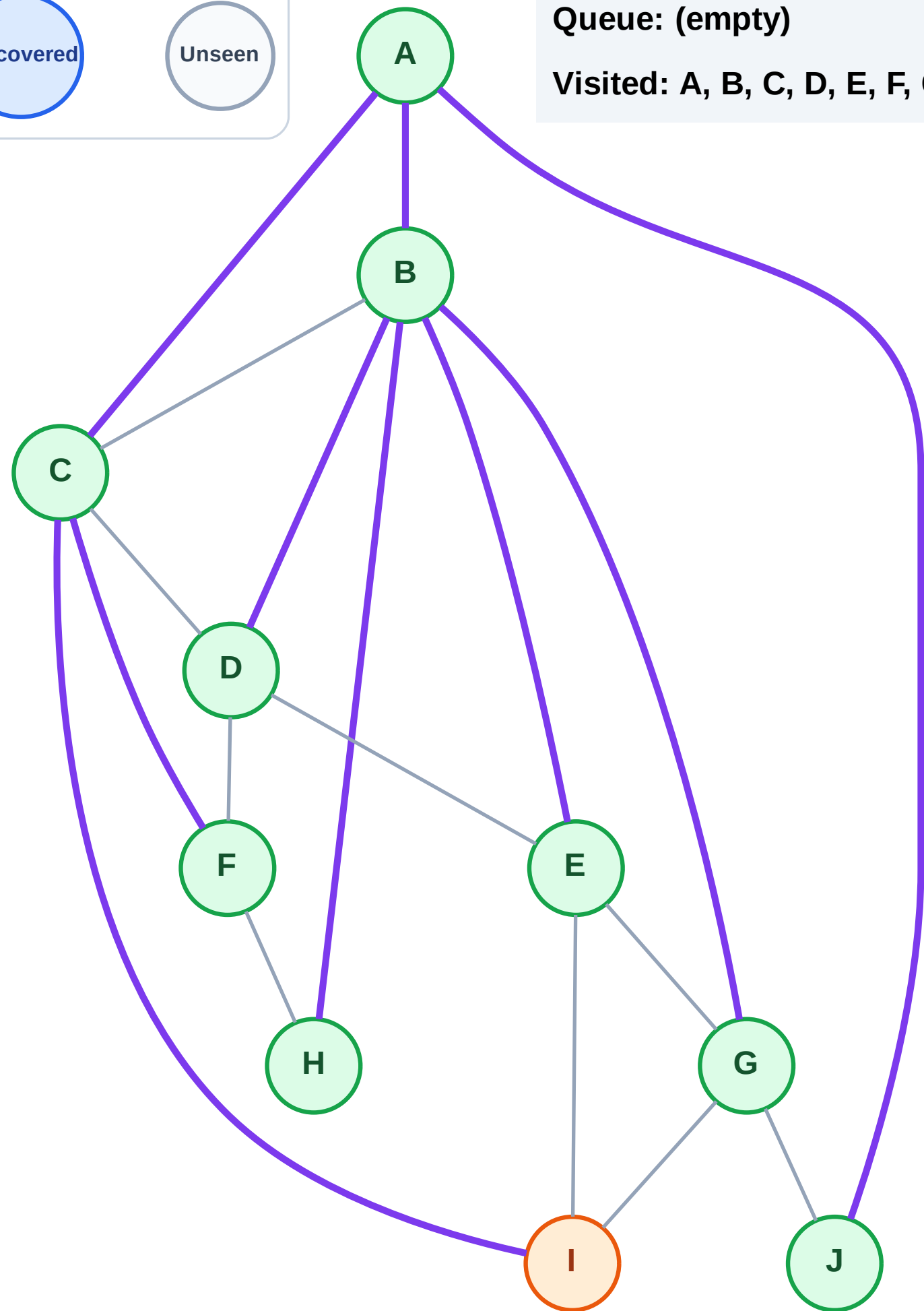
Discovered



Unseen

Queue: (empty)

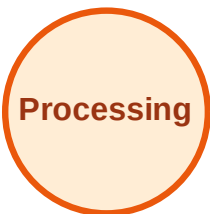
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

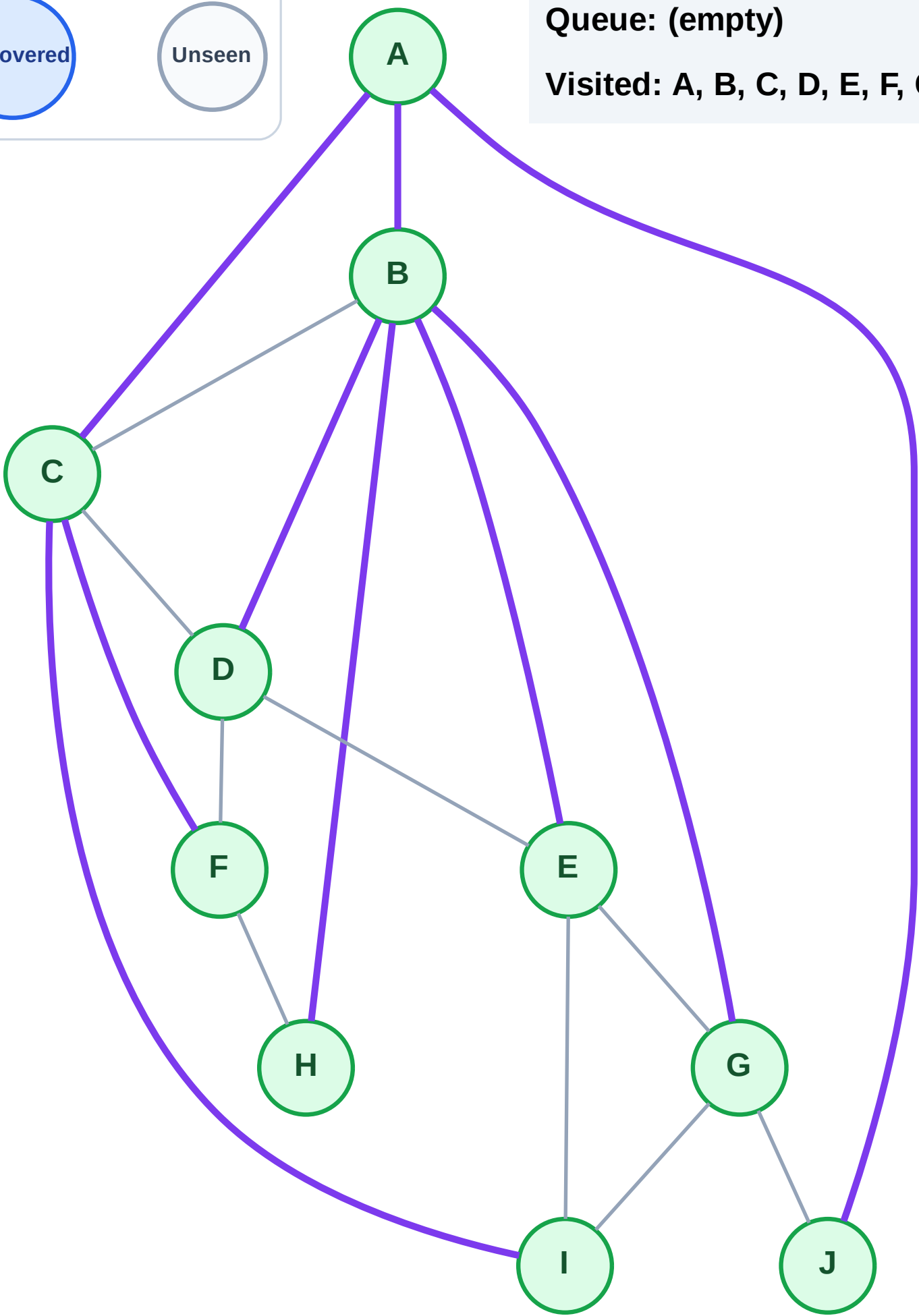
Step 21: No new neighbors from I

Legend



Queue: (empty)

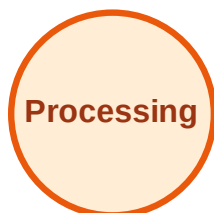
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

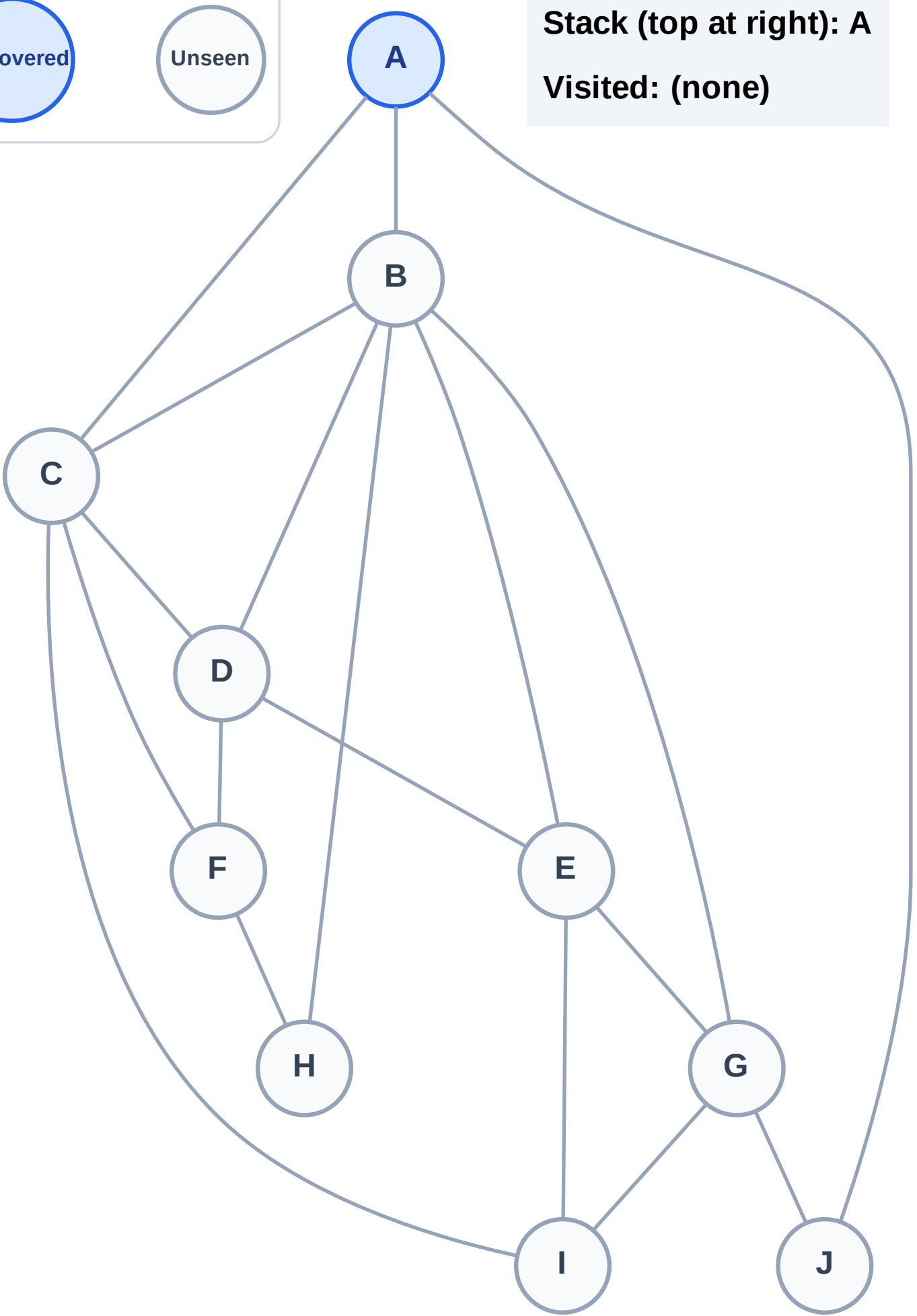
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

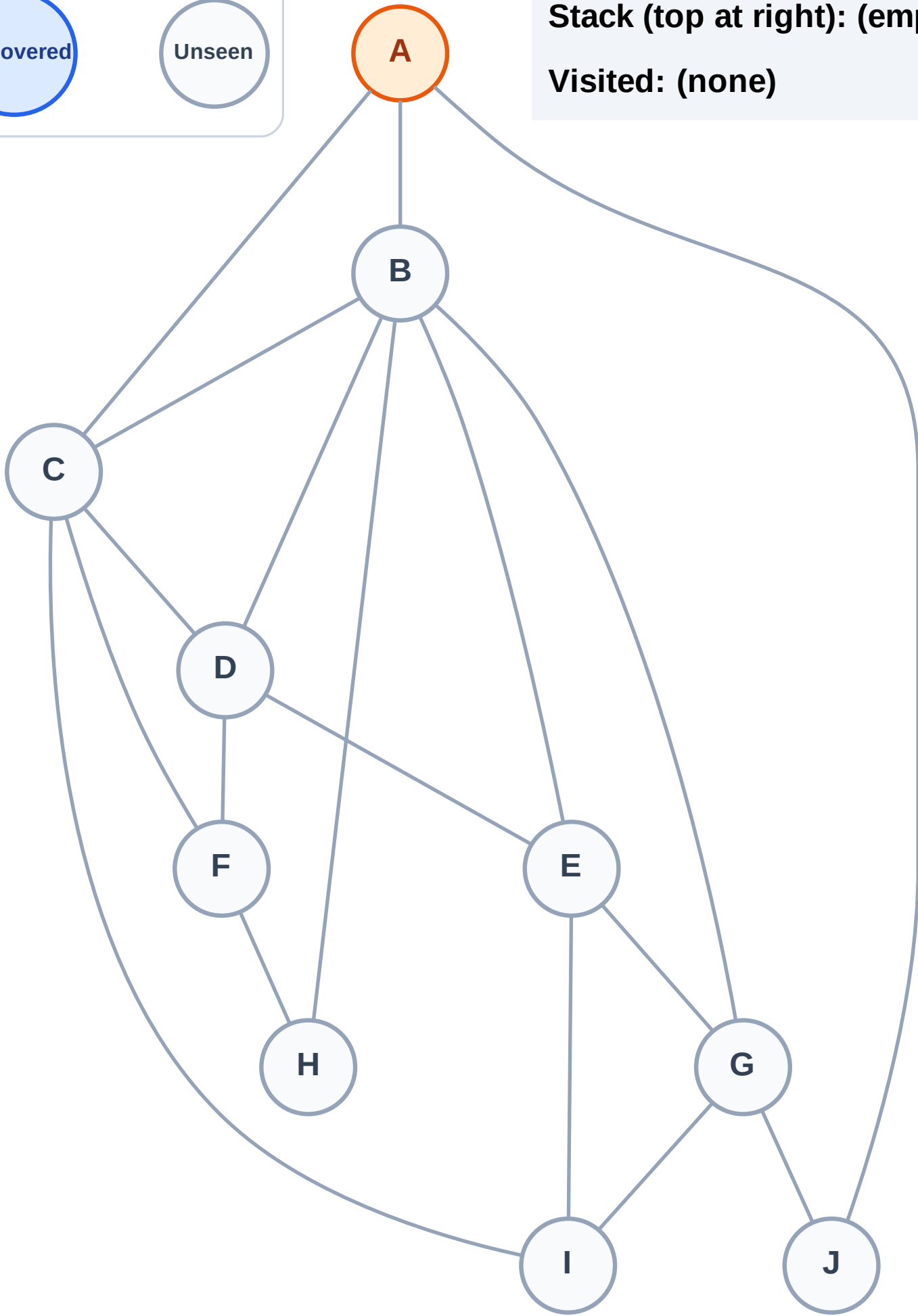
Complete

Processing

Discovered

Unseen

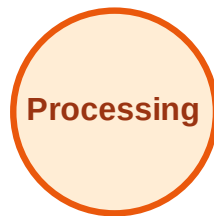
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

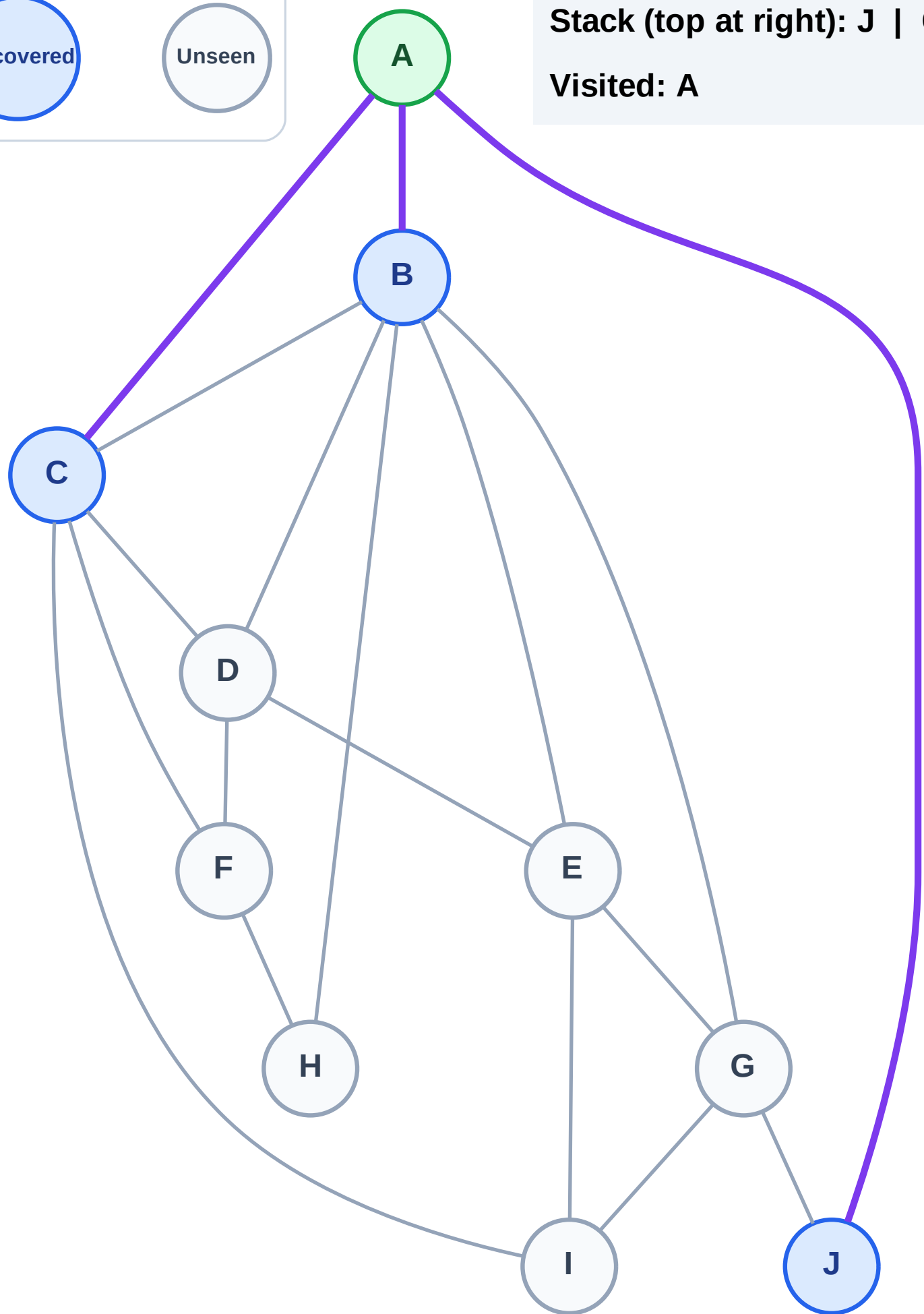
Step 3: Discover J, C, B from A

Legend



Stack (top at right): J | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

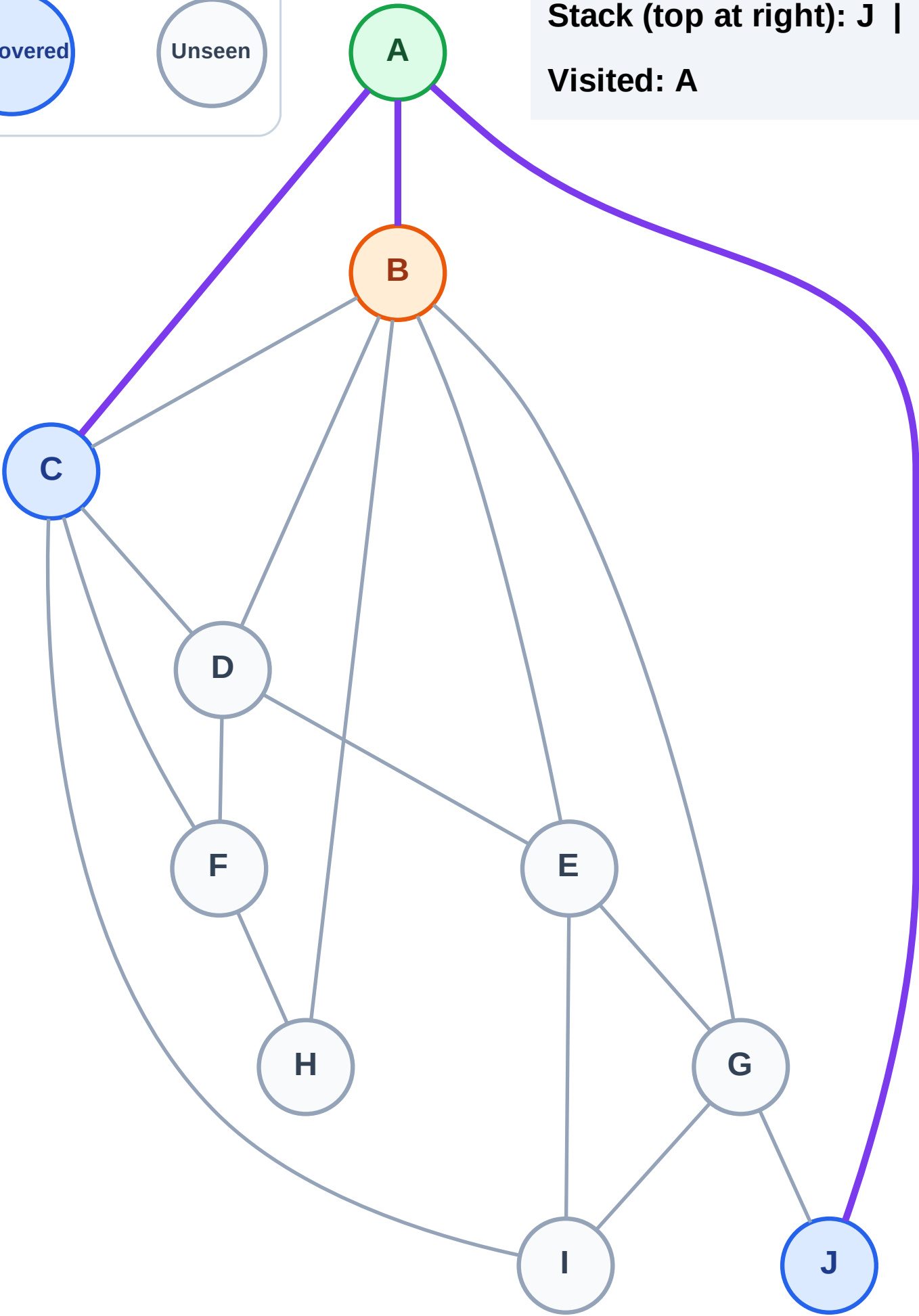
Complete

Processing

Discovered

Unseen

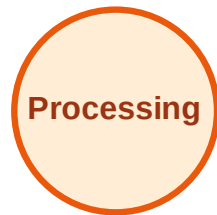
Stack (top at right): J | C
Visited: A



DFS Traversal

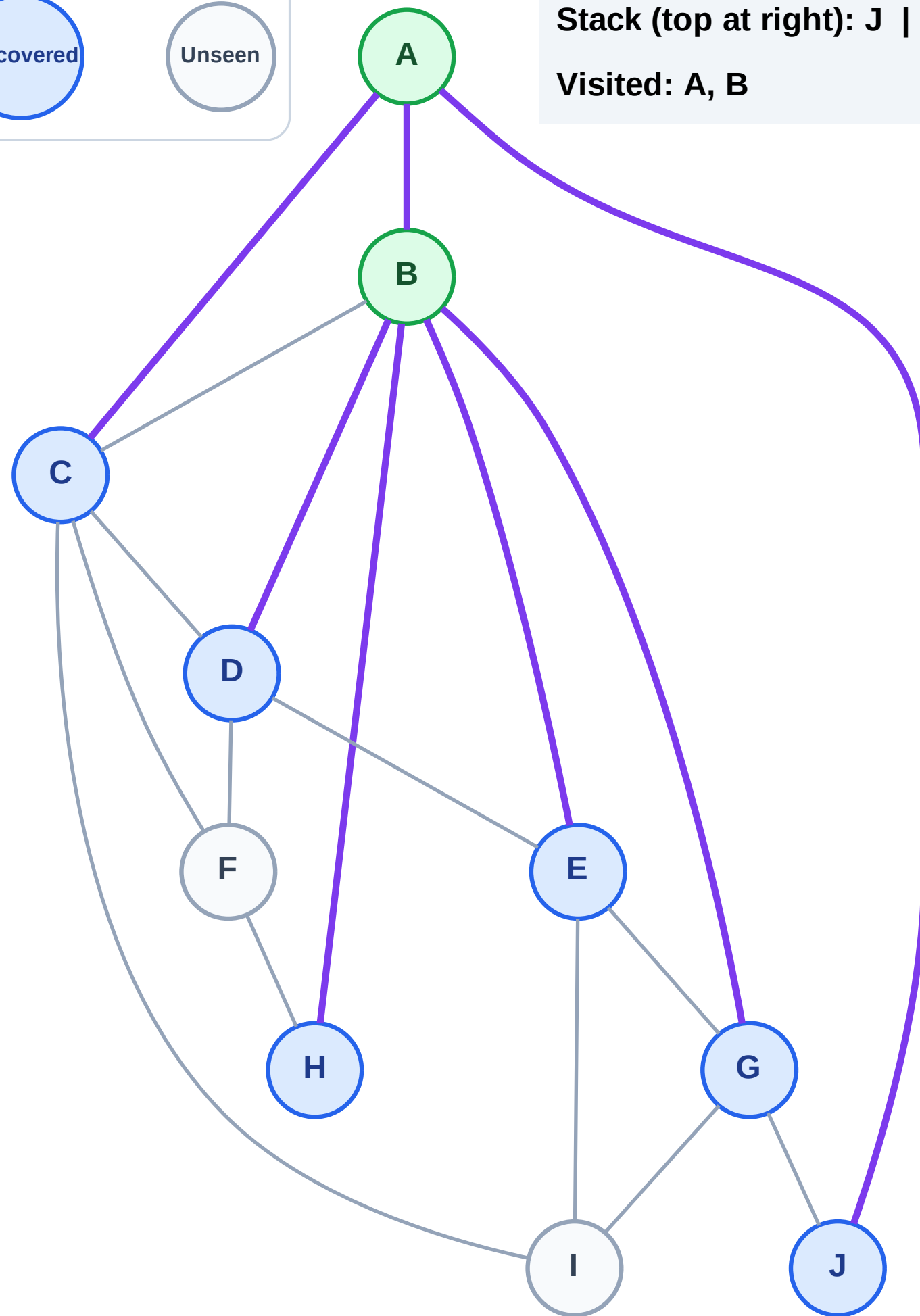
Step 5: Discover H, G, E, D from B

Legend



Stack (top at right): J | C | H | G | E | D

Visited: A, B



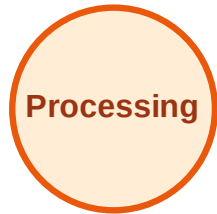
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



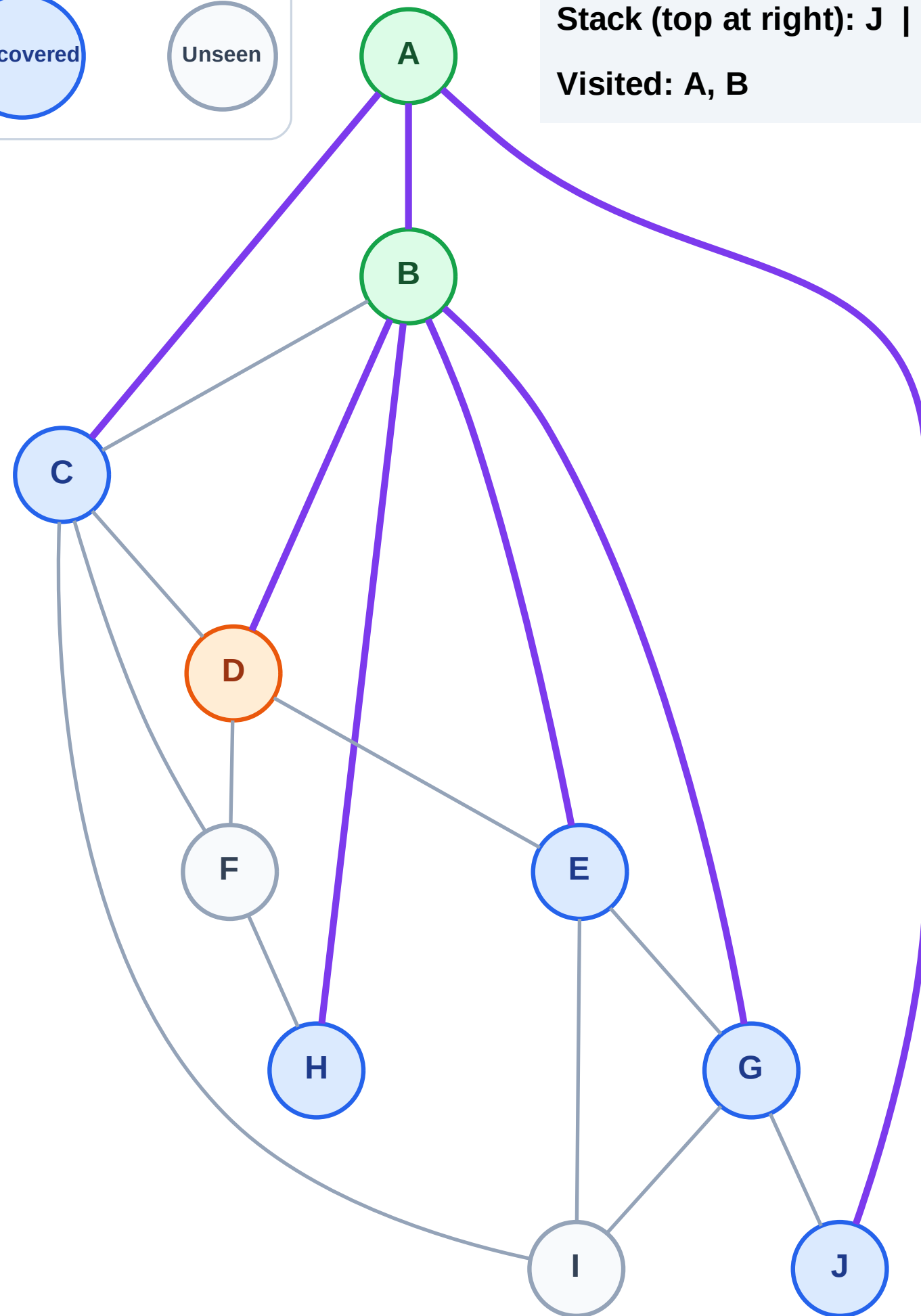
Discovered



Unseen

Stack (top at right): J | C | H | G | E

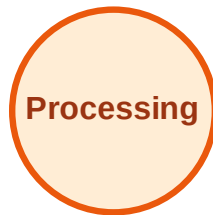
Visited: A, B



DFS Traversal

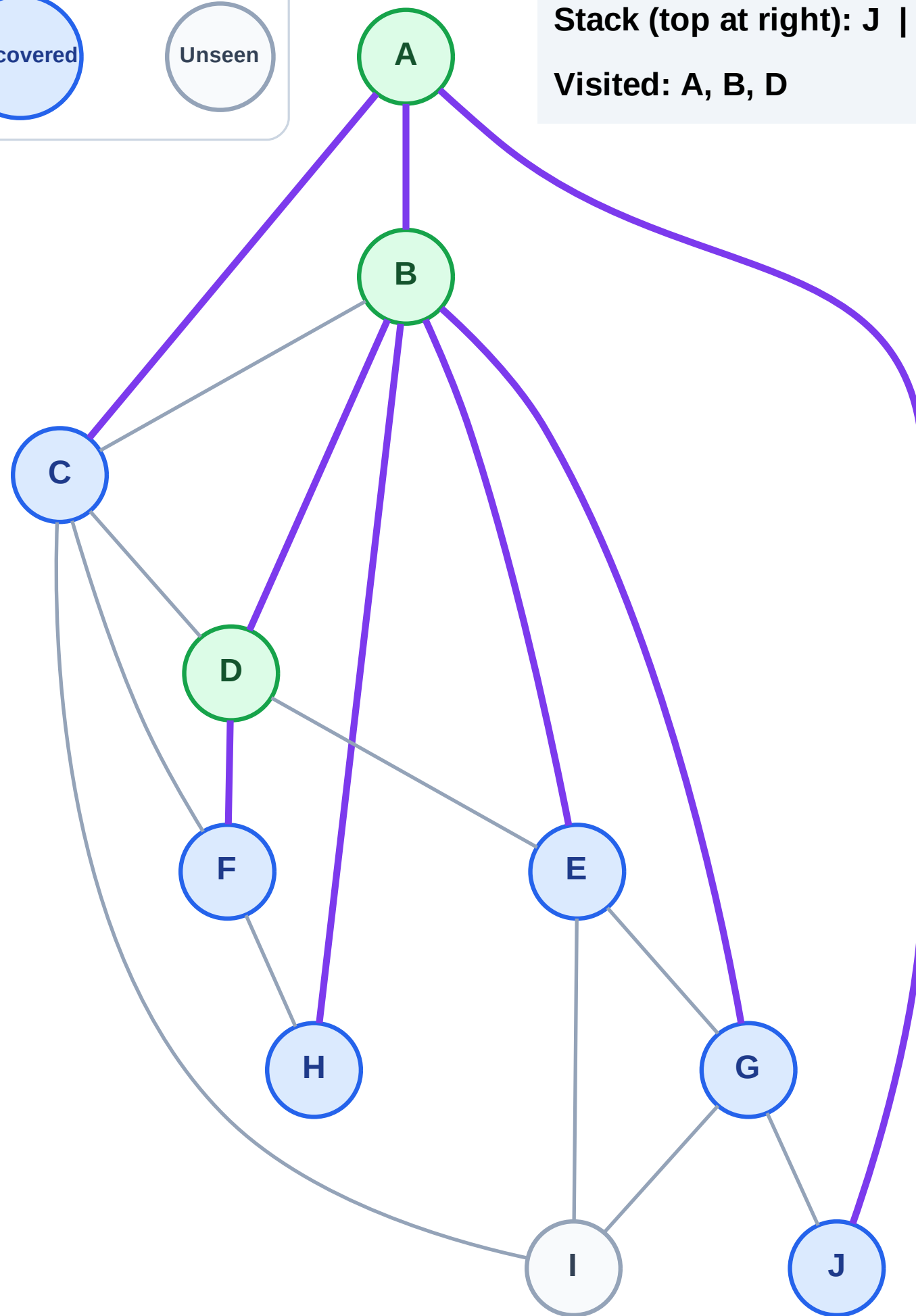
Step 7: Discover F from D

Legend



Stack (top at right): J | C | H | G | E | F

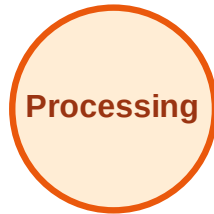
Visited: A, B, D



DFS Traversal

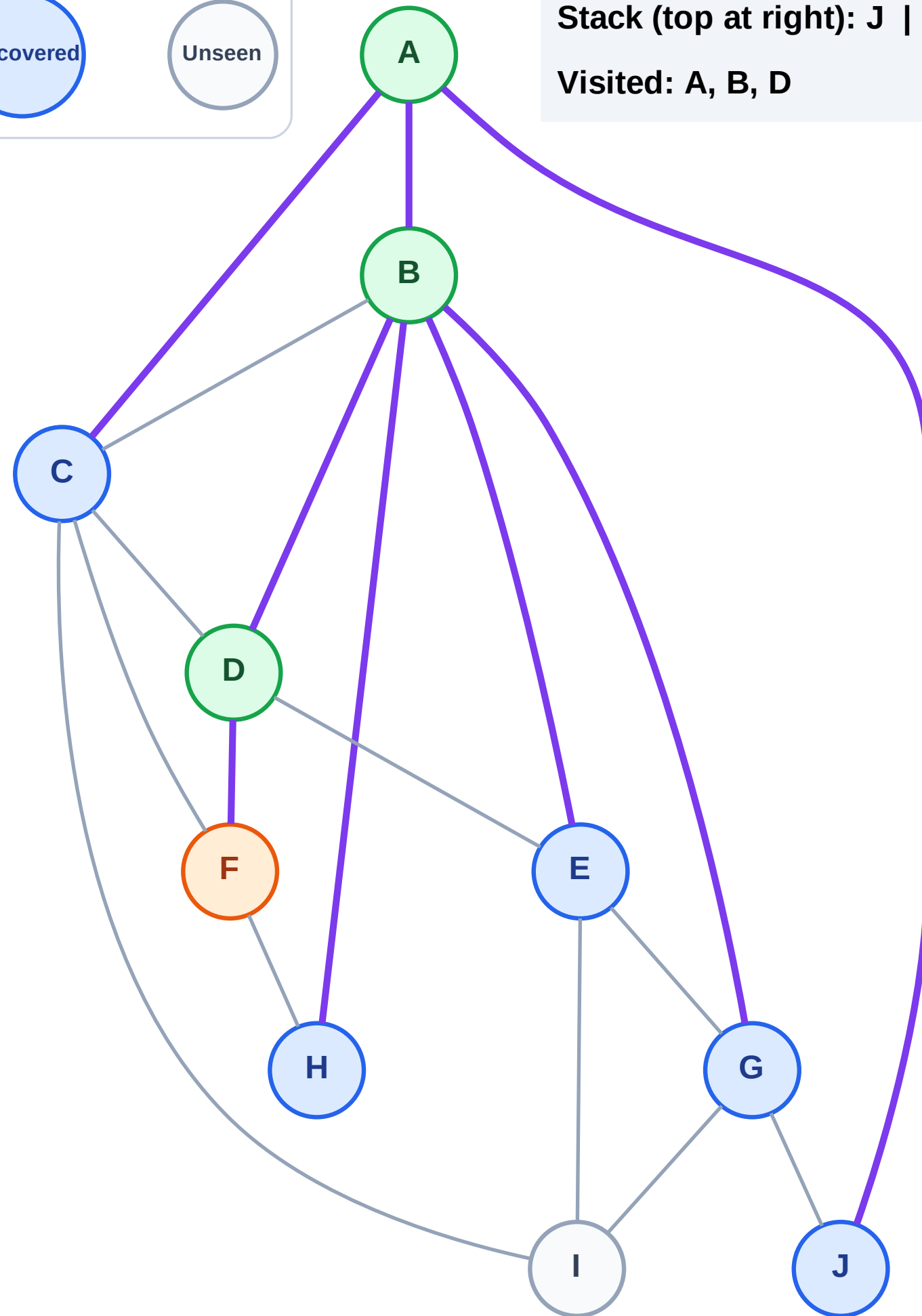
Step 8: Process node F

Legend



Stack (top at right): J | C | H | G | E

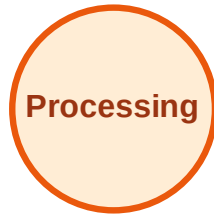
Visited: A, B, D



DFS Traversal

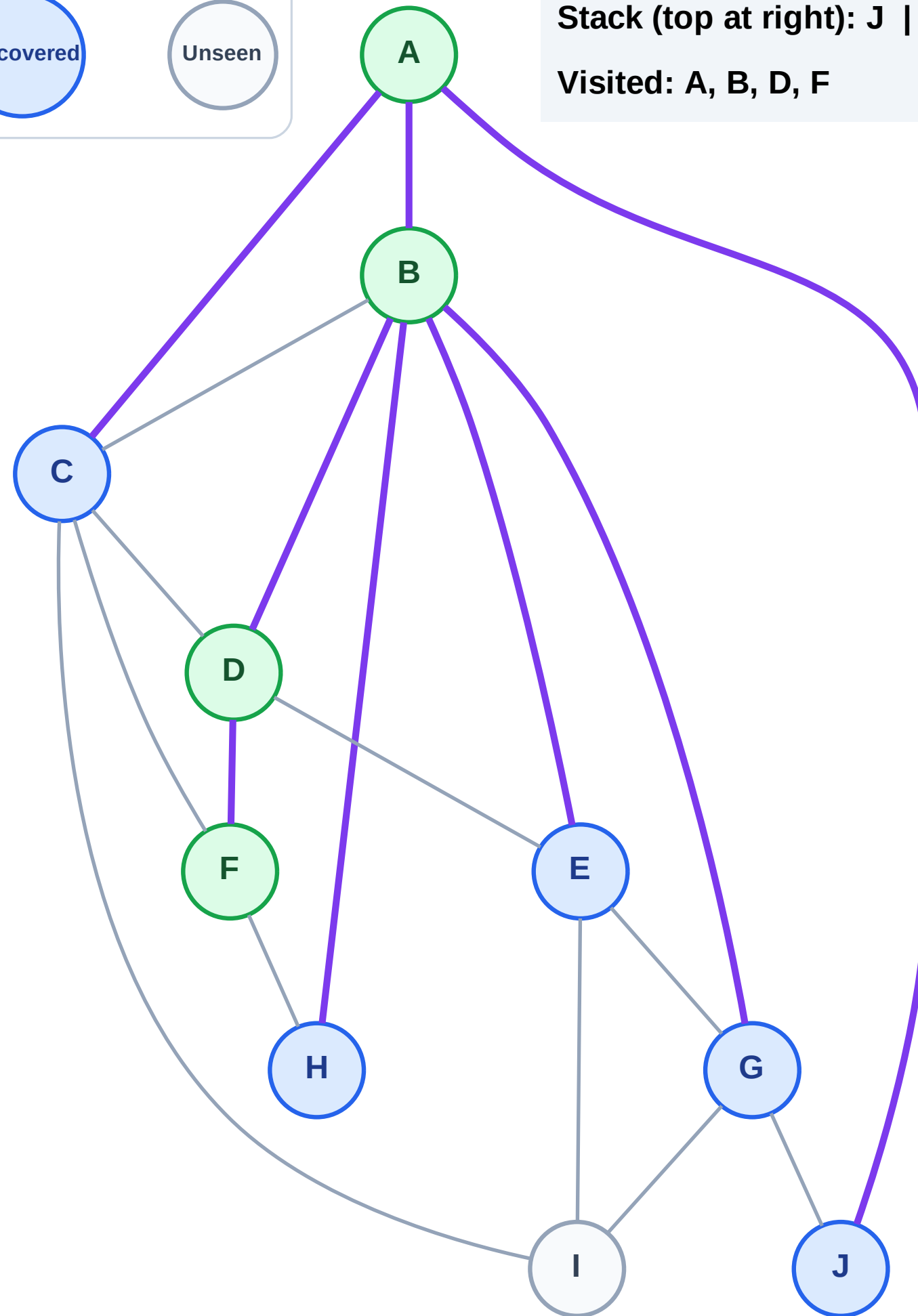
Step 9: No new neighbors from F

Legend



Stack (top at right): J | C | H | G | E

Visited: A, B, D, F



DFS Traversal

Step 10: Process node E

Legend

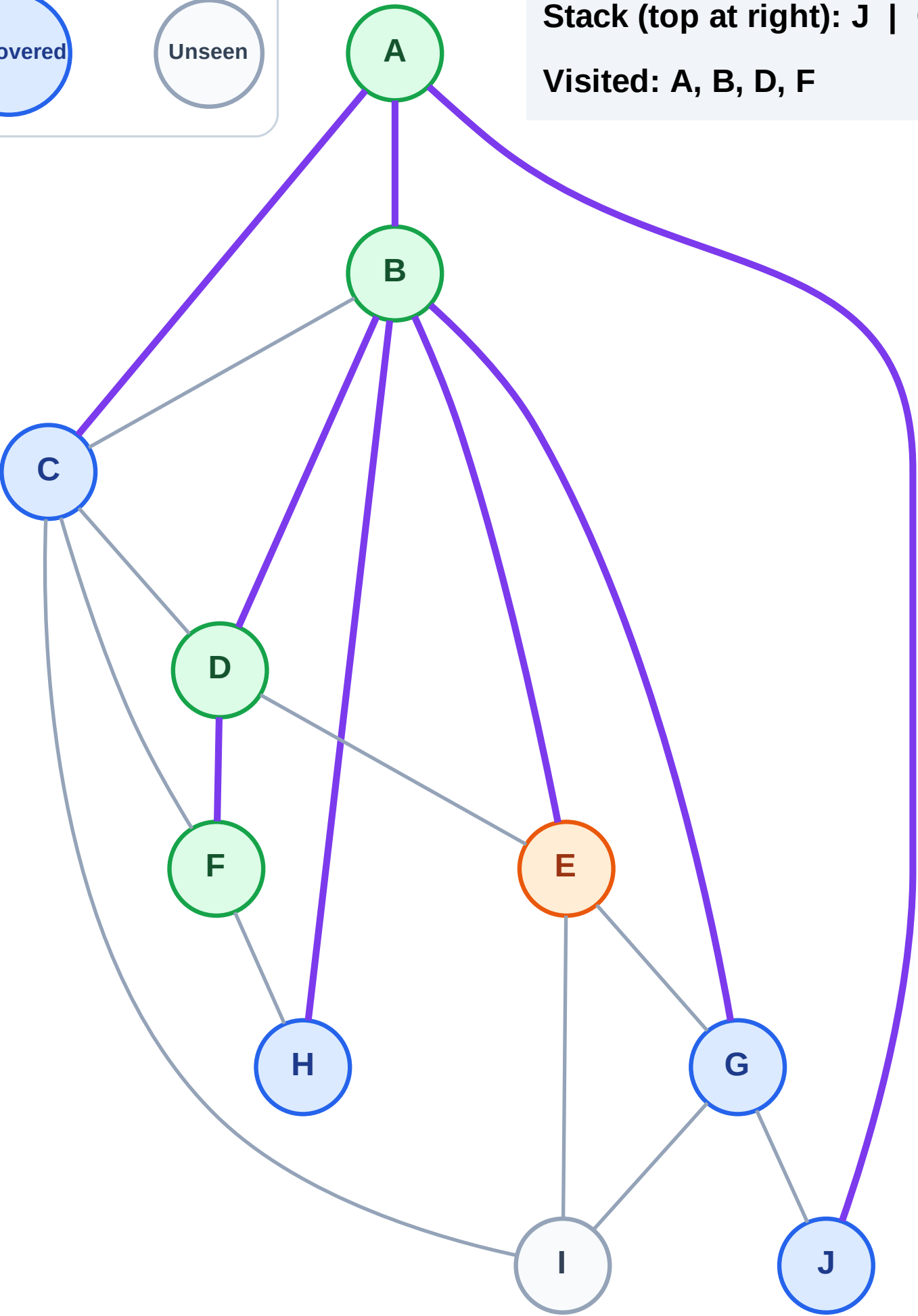
Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | H | G
Visited: A, B, D, F



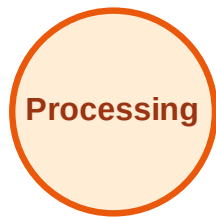
DFS Traversal

Step 11: Discover I from E

Legend



Complete



Processing



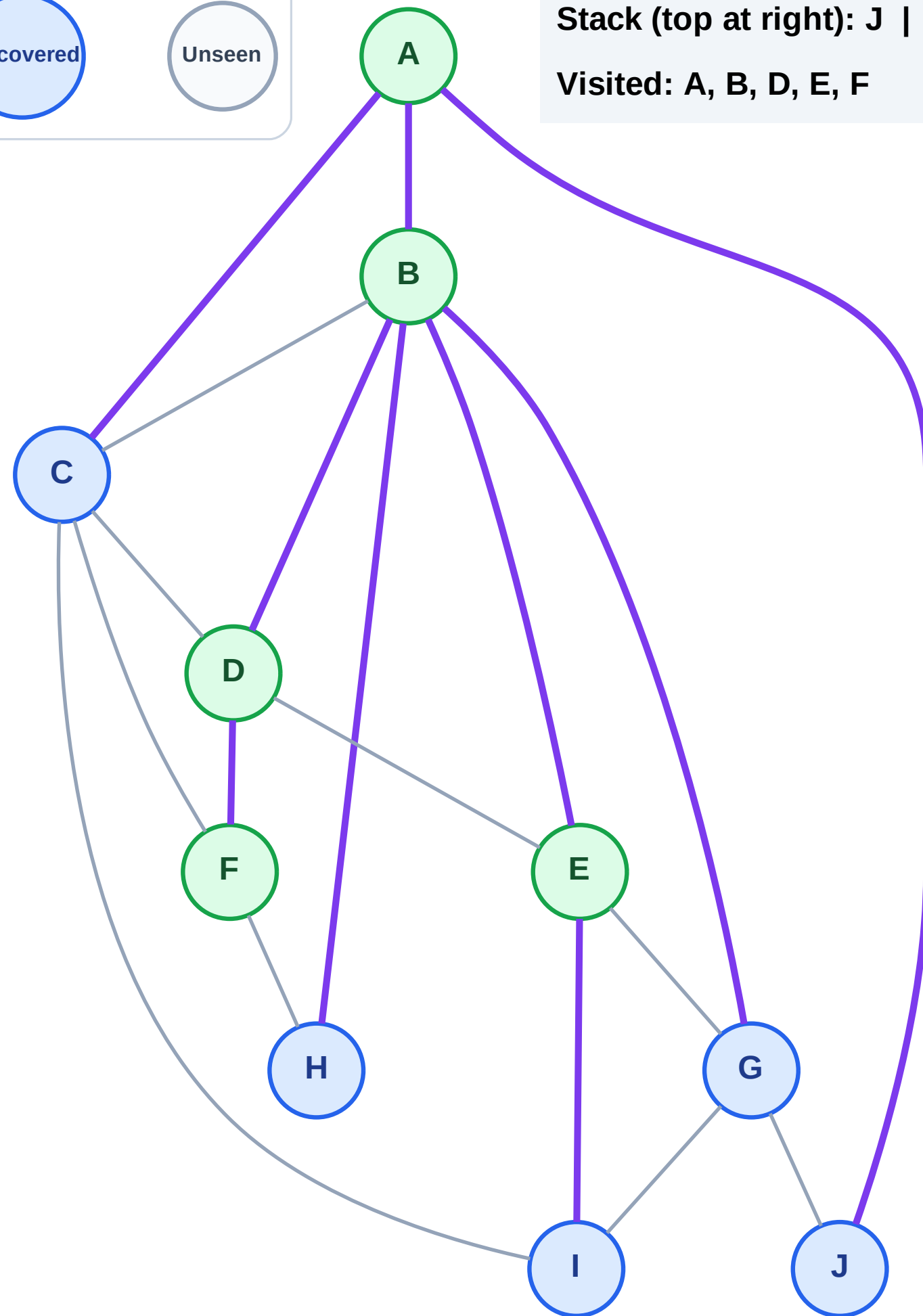
Discovered



Unseen

Stack (top at right): J | C | H | G | I

Visited: A, B, D, E, F



DFS Traversal

Step 12: Process node I

Legend

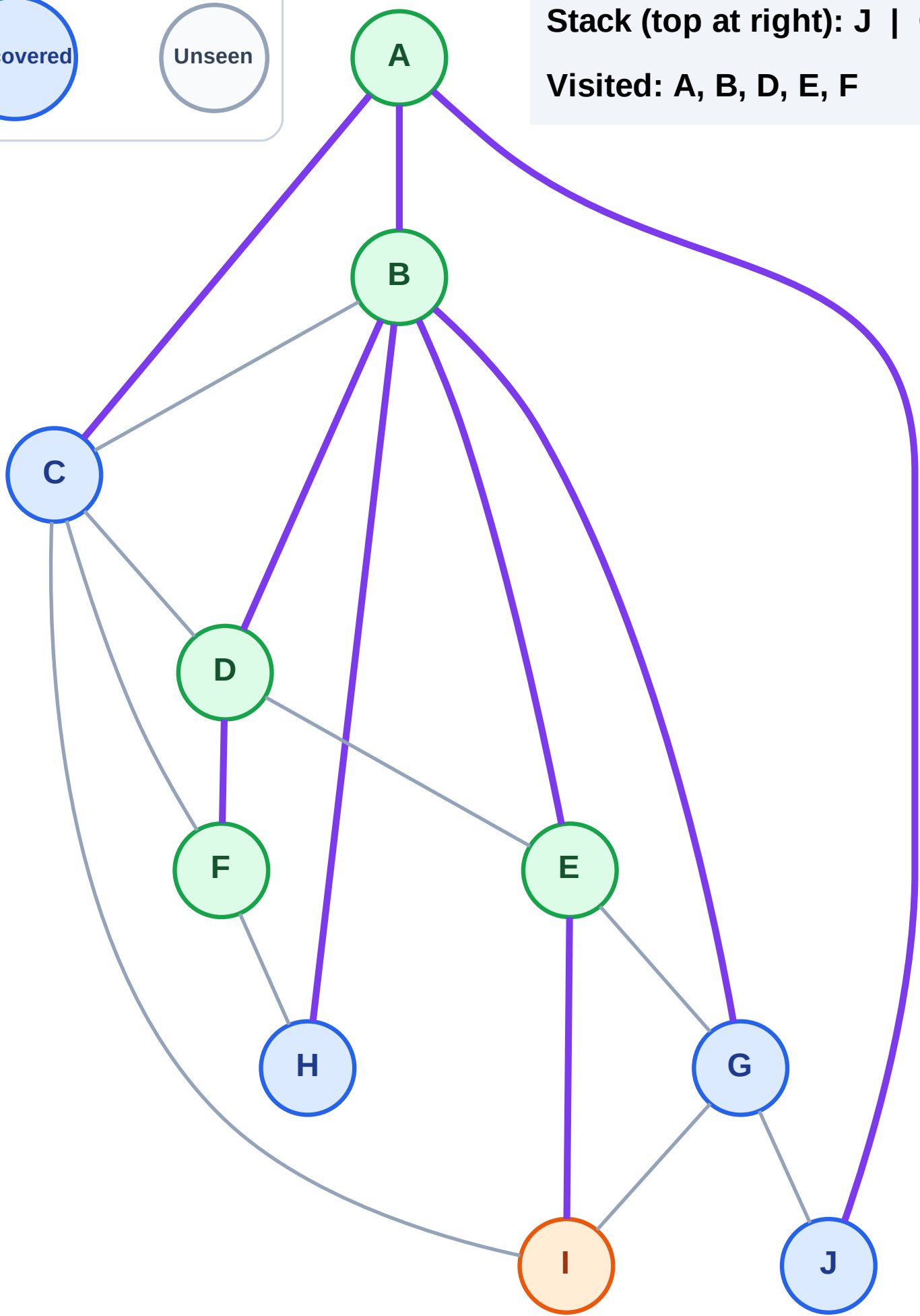
Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | H | G
Visited: A, B, D, E, F



DFS Traversal

Step 13: No new neighbors from I

Legend

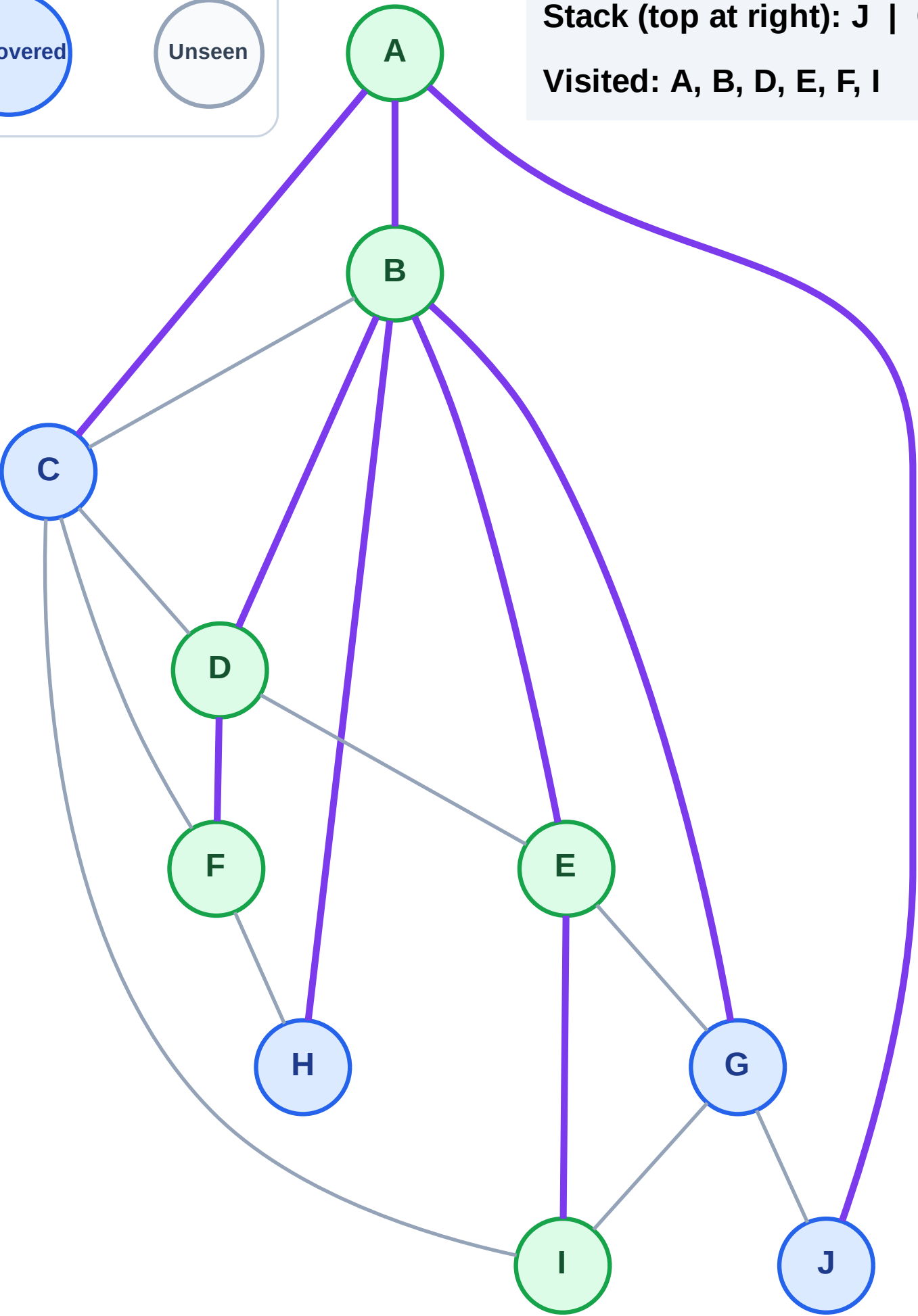
Complete

Processing

Discovered

Unseen

Stack (top at right): J | C | H | G
Visited: A, B, D, E, F, I



DFS Traversal

Step 14: Process node G

Legend

Complete

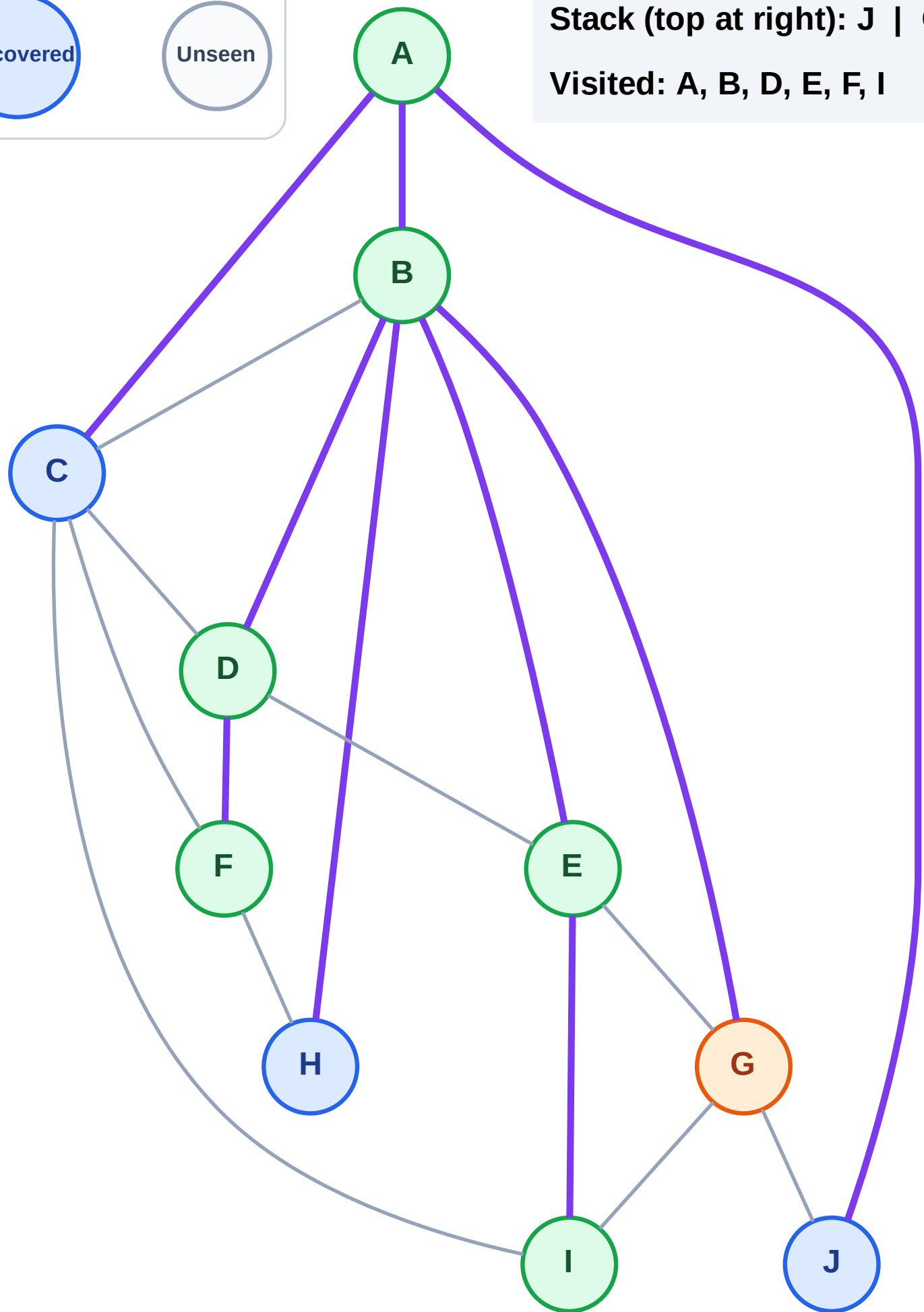
Processing

Discovered

Unseen

Stack (top at right): J | C | H

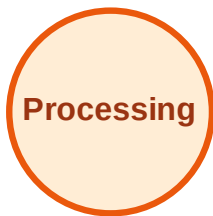
Visited: A, B, D, E, F, I



DFS Traversal

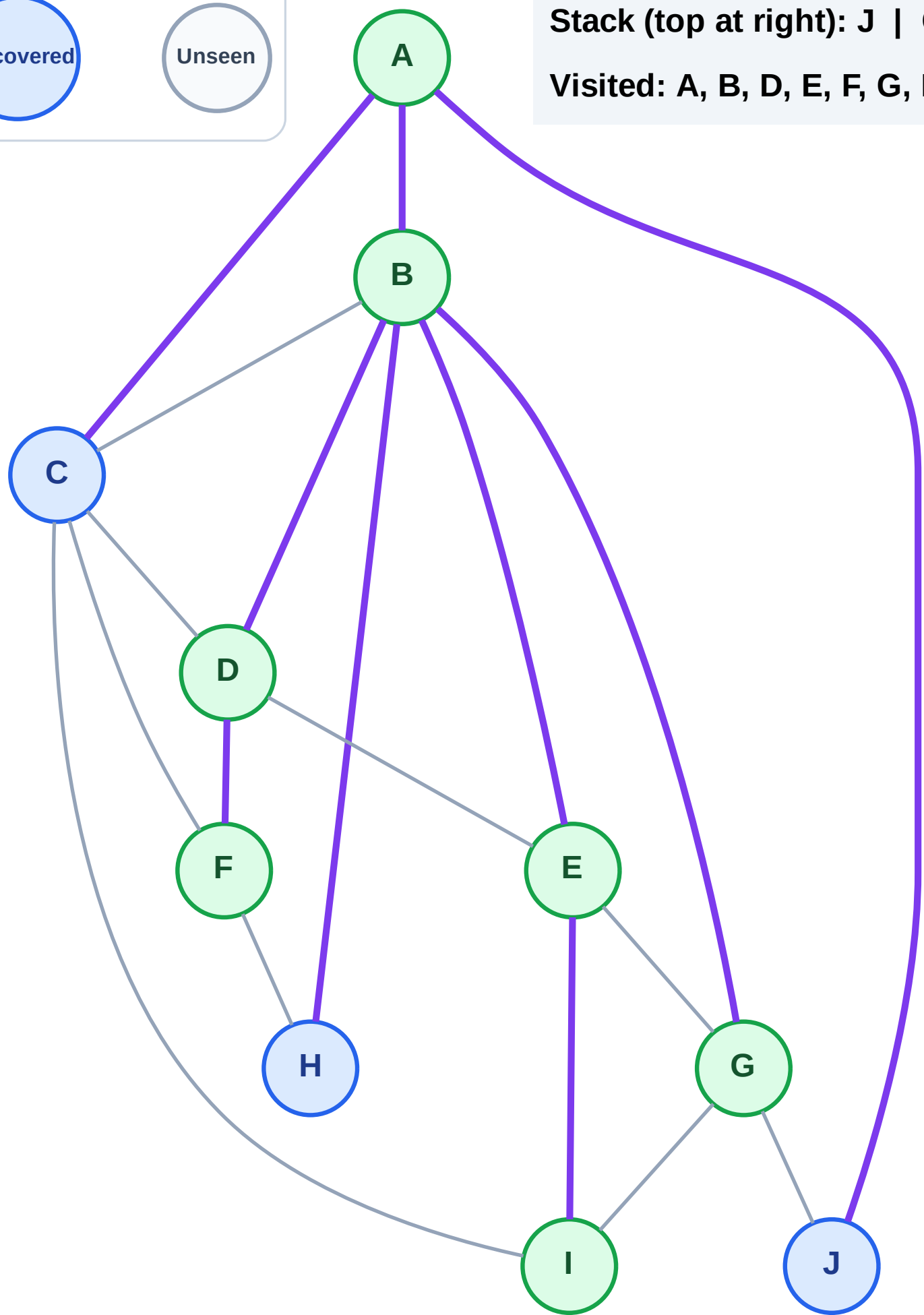
Step 15: No new neighbors from G

Legend



Stack (top at right): J | C | H

Visited: A, B, D, E, F, G, I



DFS Traversal

Step 16: Process node H

Legend

Complete

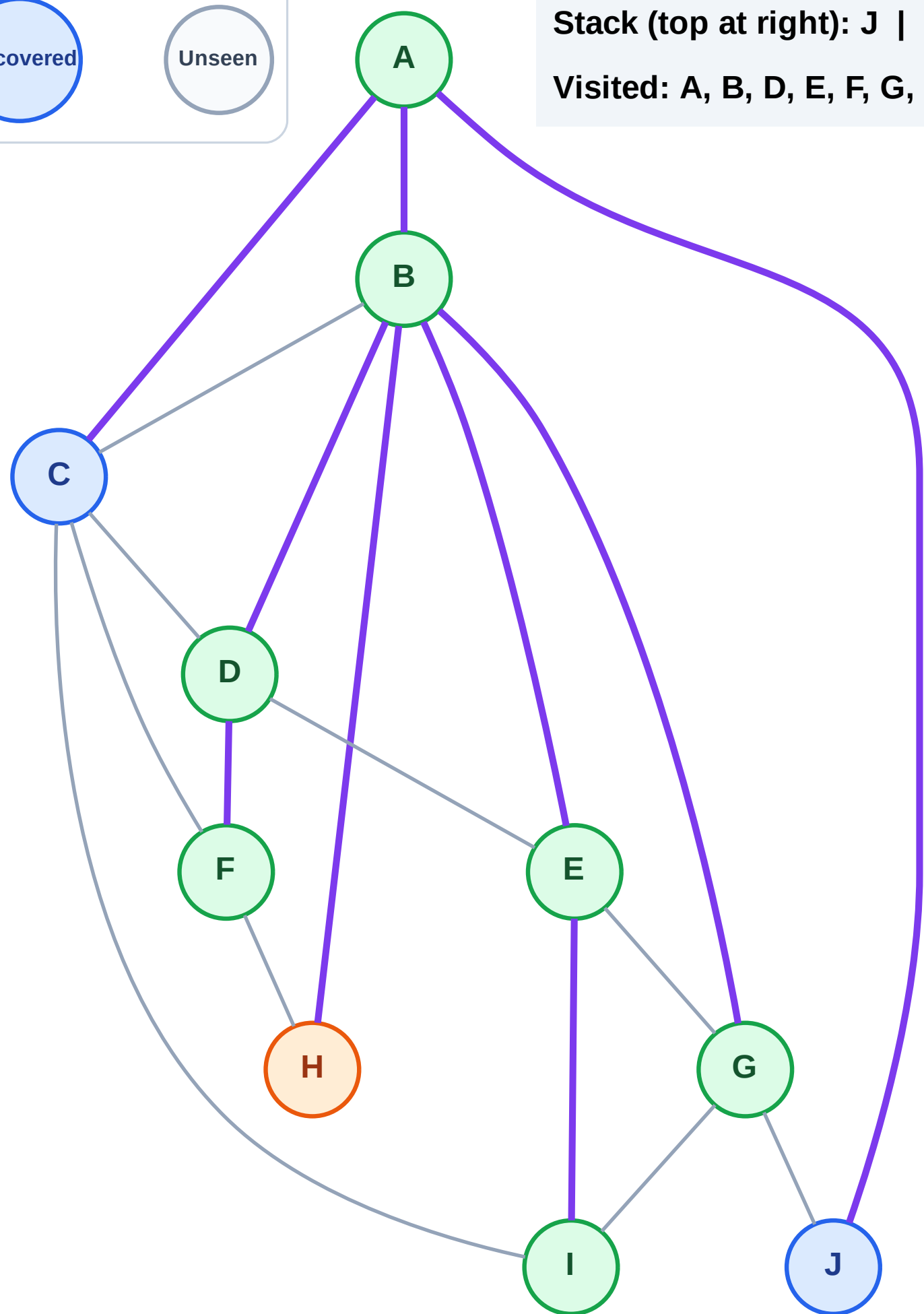
Processing

Discovered

Unseen

Stack (top at right): J | C

Visited: A, B, D, E, F, G, I



DFS Traversal

Step 17: No new neighbors from H

Legend

Complete

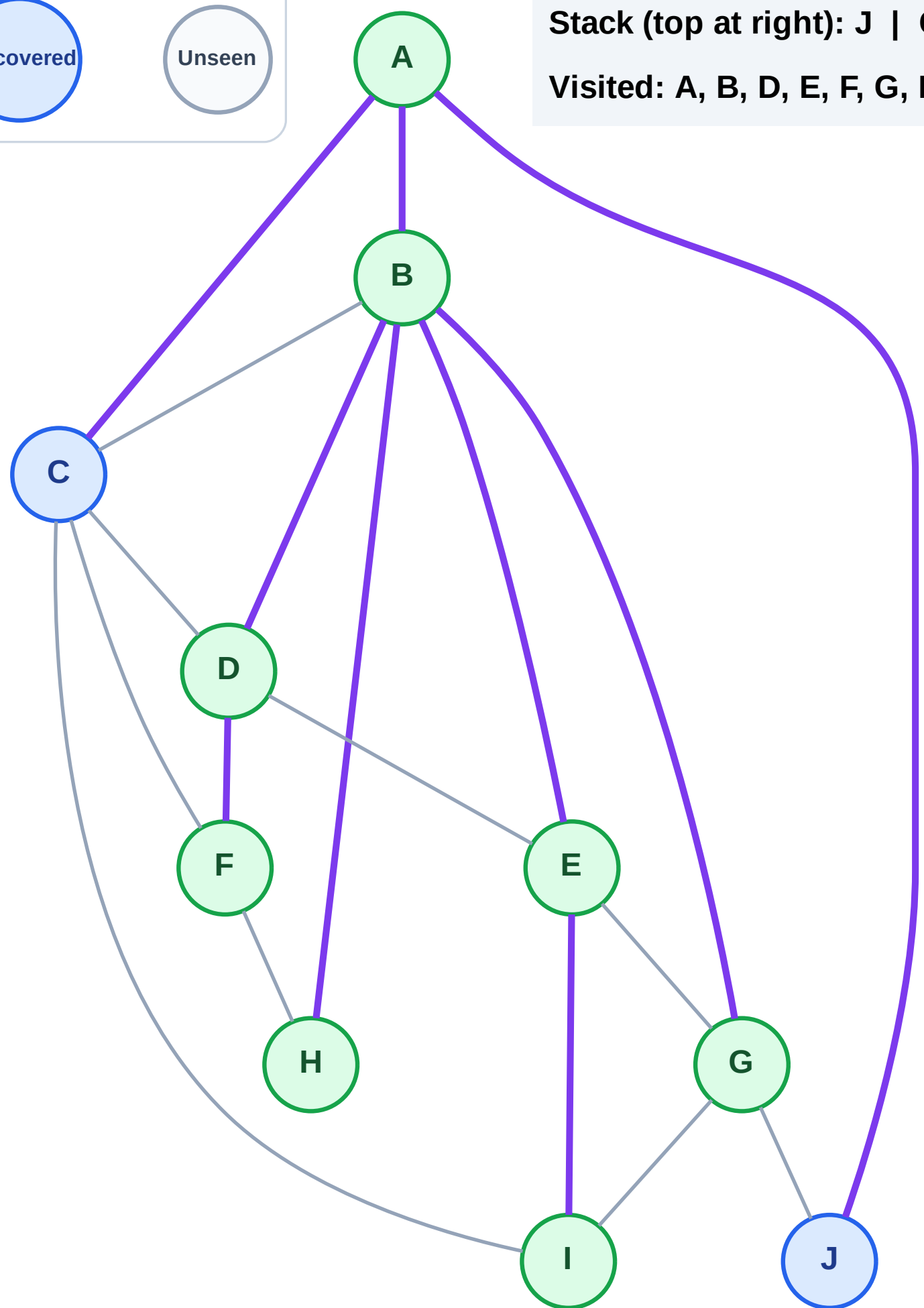
Processing

Discovered

Unseen

Stack (top at right): J | C

Visited: A, B, D, E, F, G, H, I



DFS Traversal

Step 18: Process node C

Legend

Complete

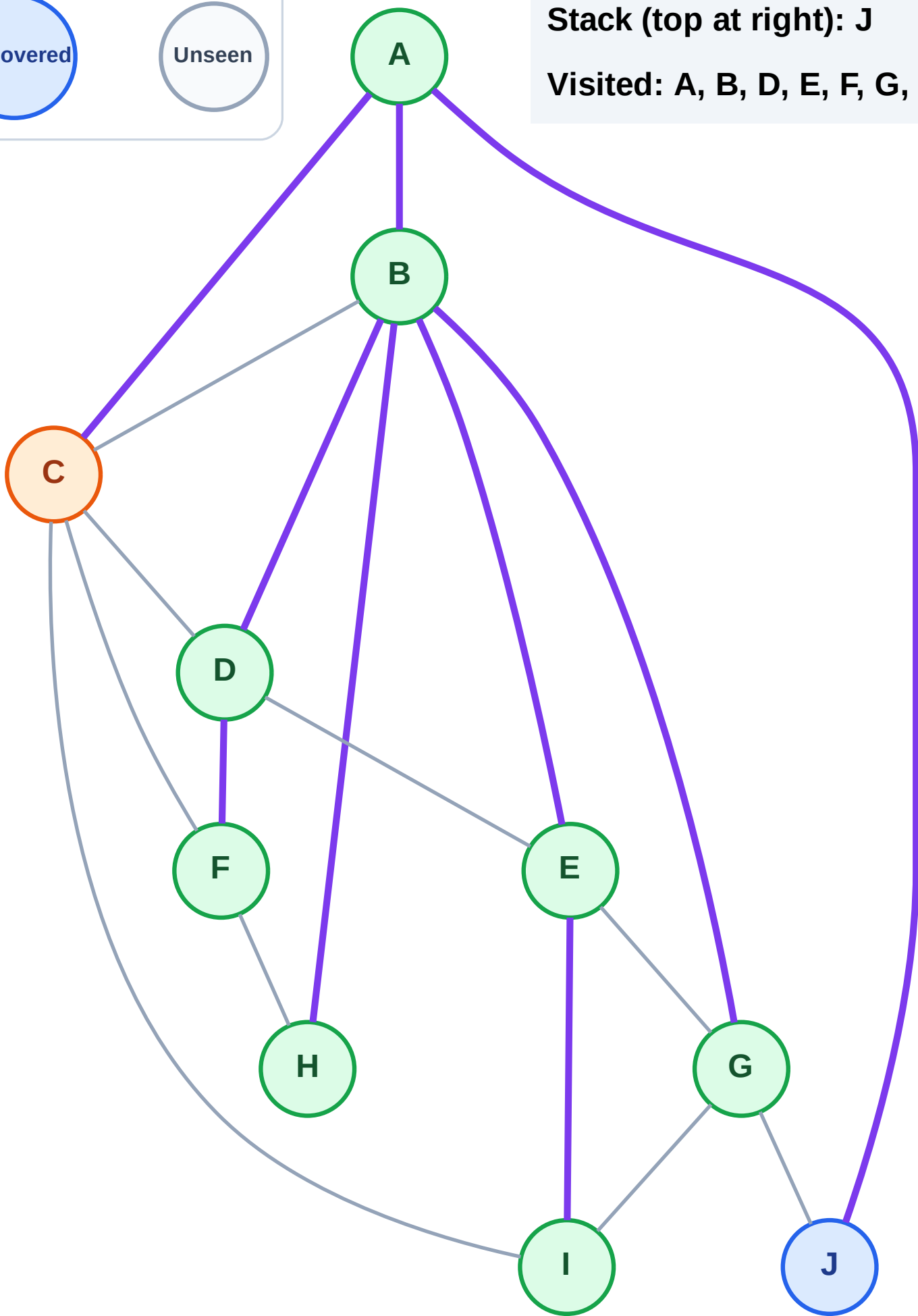
Processing

Discovered

Unseen

Stack (top at right): J

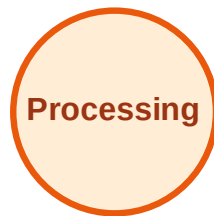
Visited: A, B, D, E, F, G, H, I



DFS Traversal

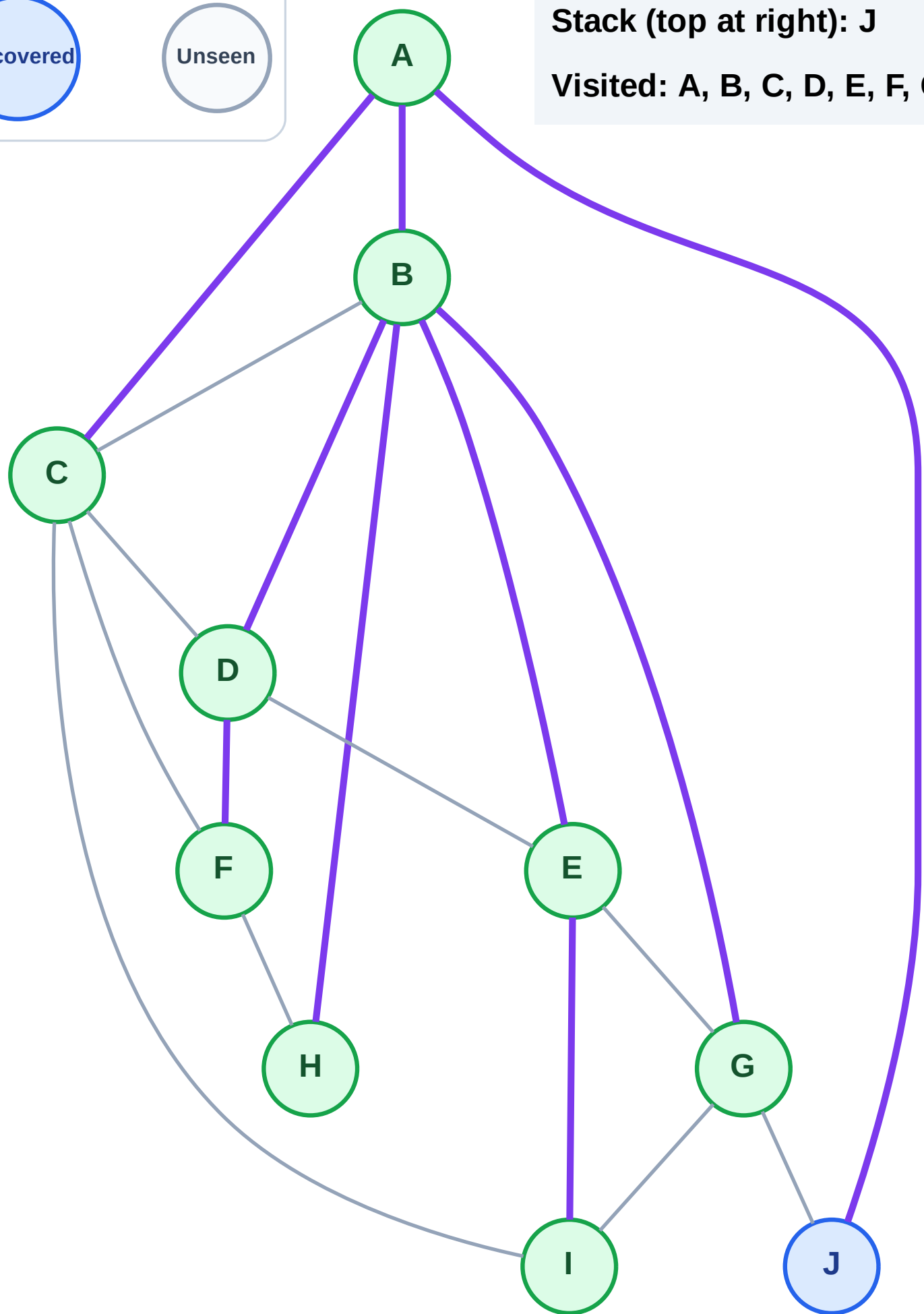
Step 19: No new neighbors from C

Legend



Stack (top at right): J

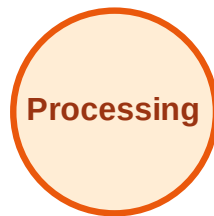
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

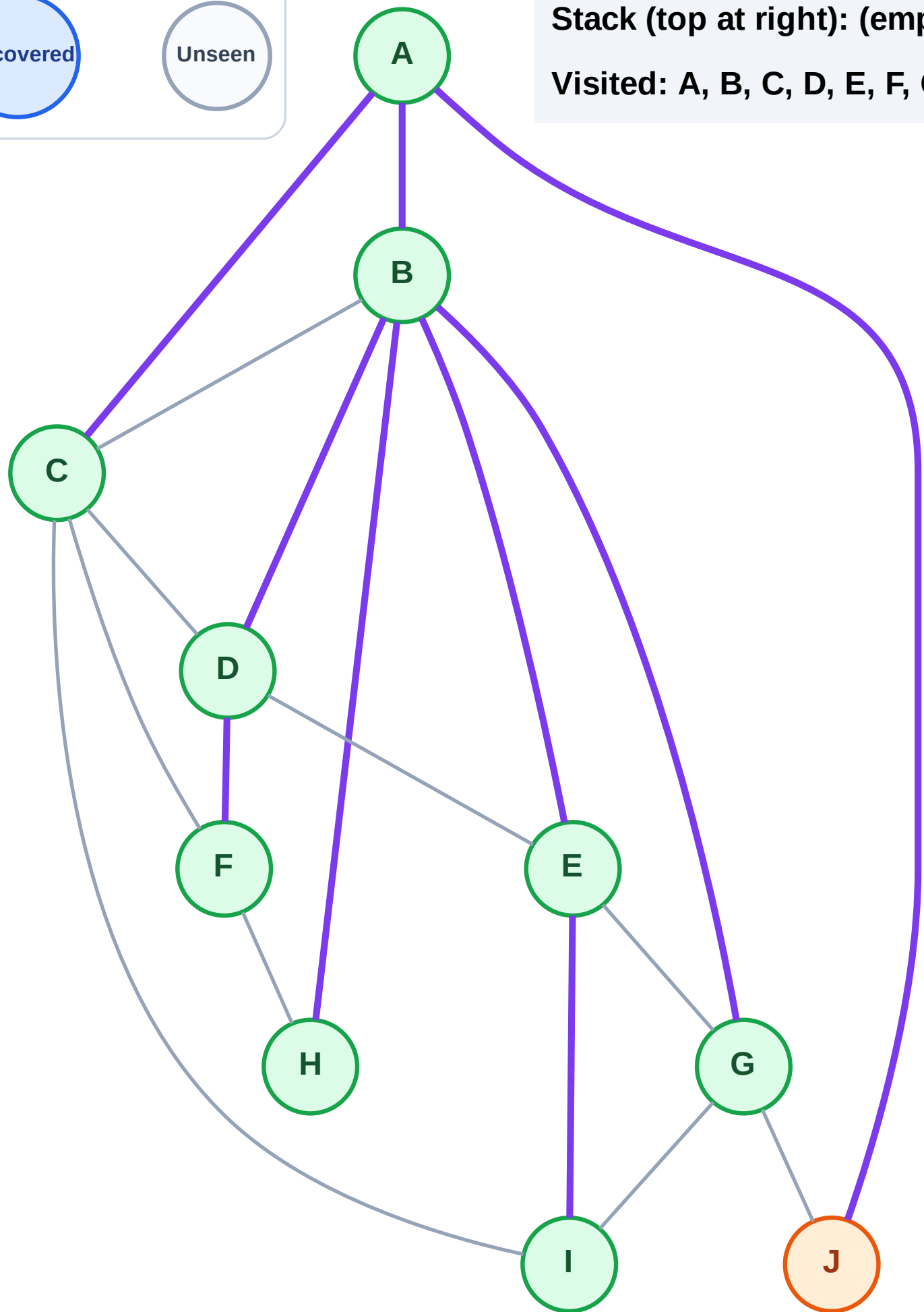
Step 20: Process node J

Legend



Stack (top at right): (empty)

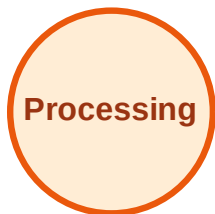
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

